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Trial-level Representational Similarity Analysis

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This study proposes a potentially **useful** improvement on a popular fMRI method for quantifying representational similarity in brain measurements by focusing on representational strength at the single trial level and adding linear mixed effects modeling for group-level inference. The manuscript provides **solid** evidence of increased sensitivity with no loss of precision compared to more classic versions of the method. However, several assumptions are insufficiently motivated, and it is unclear to what extent the approach would generalize to other paradigms.

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Abstract

Neural representation refers to the brain activity that stands in for one's cognitive experience, and in cognitive neuroscience, a prominent method of studying neural representations is representational similarity analysis (RSA). While there are several recent advances in RSA, the classic RSA (cRSA) approach examines the structure of representations across numerous items by assessing the correspondence between two representational similarity matrices (RSMs): usually one based on a theoretical model of stimulus similarity and the other based on similarity in measured neural data. However, because cRSA cannot weigh the contributions of individual trials (RSM rows/columns), it is fundamentally limited in its ability to assess subject-, stimulus-, and trial-level variances that all influence representation. Here, we formally introduce trial-level RSA (tRSA), an analytical framework that estimates the strength of neural representation for singular experimental trials and evaluates hypotheses using multi-level models. First, we verified the correspondence between tRSA and cRSA in quantifying the overall representation strength across all trials. Second, we compared the statistical inferences drawn from both approaches using simulated data that reflected a wide range of scenarios. Compared to cRSA, the multi-level framework of tRSA was both more theoretically appropriate and significantly sensitive to true effects. Third, using real fMRI datasets, we further demonstrated several issues with cRSA, to which tRSA was more robust. Finally, we presented some novel findings of neural representations that could only be assessed with tRSA and not cRSA. In summary, tRSA proves to be a robust and versatile analytical approach for cognitive neuroscience and beyond.

Introduction

Since the start of any systematic examination of the mind, the concept of *representation* has provided a link between the external world and the content of mental life (Brentano, 1874). Current perspectives in cognitive science have defined representations as brain activity patterns that convey some behaviorally relevant content, which could be sensory perception, memory, concept knowledge, or social relations (Kriegeskorte & Diedrichsen, 2019). One of the central approaches for evaluating information represented in the brain is representational similarity analysis (RSA), an analytical approach that queries the representational geometry of the brain in terms of its alignment with the representational geometry of some cognitive model (Kriegeskorte et al., 2008; Kriegeskorte & Kievit, 2013), or, in some cases, compares the representational geometry of two neural systems (e.g., Kriegeskorte et al., 2008) or two model systems (Sucholutsky et al., 2023). The RSA approach has demonstrated utility across many domains of cognitive neuroscience research, such as visual perception (Jozranjbar et al., 2023), episodic memory (Xue, 2018), concept knowledge (Bauer & Just, 2019), social information (Freeman et al., 2018), and cognitive control (Freund, Etzel, et al., 2021). Despite its proven success, classic RSA approach, henceforth cRSA, has several limitations in efficacy across various experimental scenarios, due to its inability to reflect the proper multi-level variance structure within the data. Recently, methodological advancements have addressed many known limitations in cRSA. For example, cross-validated distance measures have improved the reliability of representational dissimilarities in the presence of noise and trial imbalance (Diedrichsen et al., 2021; Nili et al., 2014; Walther et al., 2016). Bayesian approaches such as pattern component modeling (Diedrichsen et al., 2018) have extended representational approaches to accommodate continuous stimulus features or temporal variation. Further, model comparison RSA strategies (Diedrichsen et al., 2021) and generalization techniques across stimuli and participants (Schütt et al., 2023) have improved sensitivity and inference. Nevertheless, a common feature shared across most of improvements is that they require stimuli repetition to examine the representational structure. This requirement limits their ability to probe brain-behavior questions at the level of individual events. In this paper, we present an advancement termed trial-level RSA, or tRSA, which addresses these limitations in cRSA and may be utilized in paradigms with or without repeated stimuli.

A typical implementation of cRSA involves four main steps (see Figure 1). First, brain activity responses to a series of N trials are compared against each other (typically using Pearson's r) to form an $N \times N$ representational similarity matrix, or RSM_{brain}. Second, a hypothesis of how this brain system ought to respond — a cognitive model — is created in the form of a model RSM, or RSM_{model}. This RSM_{model} can be created based on objective features of the stimuli (e.g., image brightness, category membership), subject ratings or behaviors (e.g., pleasantness, memory success), or outputs from computational models (e.g., neuron activations in artificial neural networks, semantic embeddings from large language models). Third, values from the lower (or equivalently, upper) triangular parts of both RSM_{brain} and RSM_{model} are retrieved, vectorized, and compared. This RSM_{brain}-RSM_{model} comparison focuses on the similarity between the two representational geometries, which is typically measured by Spearman's rank correlation (Kriegeskorte et al., 2008) but alternatives have also been proposed (Bobadilla-Suarez et al., 2020; Diedrichsen et al., 2021; Walther et al., 2016). This similarity measure is often referred to as a first-level RSA score or *representational strength* (*representational geometry* usually refers to the structure of similarities among repeated presentations of the same stimulus in the neural data [as captured in the brain RSM] and is often estimated utilizing a model comparison approach, whereas *representational strength* is a derived measure that quantifies how strongly this geometry aligns with a hypothesized model RSM; in other words, geometry characterizes the pattern space itself, while representational strength reflects the degree of correspondence between that space and the theoretical model under test). Fourth, once the above procedures are completed for each subject and condition, the first-level measures of representation are submitted to a second-level hypothesis testing for statistical inferences — often using general linear models such as t-test and ANOVA. One notable feature of cRSA is that the representational geometries are compared to one another in their entireties in step 3, and this step produces as a single measure of representational strength collapsing across all experimental trials. In other words, cRSA estimates cannot be interpreted for specific experimental trials or stimuli; rather, they only reflect how strongly the studied neural system represents for the entire set of items, limiting the realm of possible research questions. Most critically, as trial-level information is collapsed in cRSA, multiple sources of variance become intractable, yet these ignored variances could render subsequent second-level analyses susceptible to erroneous inferences.

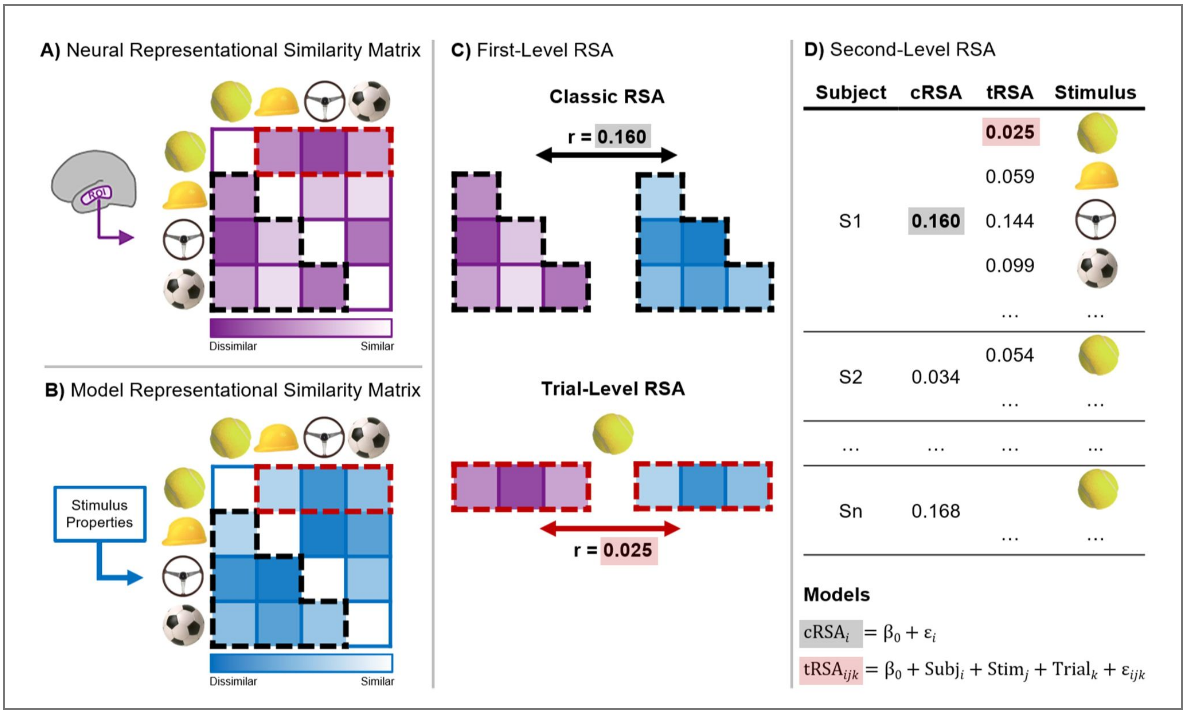


Figure 1. Representational Similarity Analysis (RSA).

Steps for classic RSA (cRSA) and trial-level RSA (tRSA). **A)** Neural Representational Similarity Matrix (RSM_{brain}) is generated by correlating multi-voxel activity patterns across all trials within a Region of Interest (ROI), reflecting similarity between neural responses. **B)** Model Representational Similarity Matrix (RSM_{model}) is constructed by correlating of-interest stimulus properties across all trials. **C)** First-Level cRSA (top) and tRSA (bottom). For cRSA, the lower triangular parts (black outline) of RSM_{brain} and RSM_{model} are compared, producing a single summary statistic (e.g., Spearman's rho) across all trials. For tRSA, representational similarity values from RSM_{brain} and RSM_{model} for the same trial (e.g., tennis ball; red dashed outline) are compared, producing a single representational strength estimate for that trial. **D)** Second-Level analysis for cRSA and tRSA. In cRSA, subject-level r values are submitted to a one-sample t-test to assess whether the values reliably exceed zero. In tRSA, a linear random effects model with random effects for subject and stimulus is fit, and hypothesis testing determines whether the estimated intercept is significantly greater than zero.

A behavioral or neural measure in a single experimental trial consists of meaningful variances from four distinct sources: *condition-level* (experimental task manipulations such as cognitive load and emotional valence), *subject-level* (individual differences in perceptual acuity, prior knowledge, or other cognitive faculties), *stimulus-level* (features of the stimuli with behavioral or neural relevance), and *trial-level* (physiological and nuisance variables affecting measurement). Notably, the distinction between condition-level and stimulus level is not always clear as researchers may manipulate stimulus-level features themselves. In these cases, what researchers ultimately consider condition-level and stimulus-level will depend on their specific research questions. For example, researchers intending to study generalized object representation may consider object category a stimulus-level feature, while researchers interested in if/how object representation varies by category may consider the same category variable condition-level.

To explain the variance in behavioral or neural measures, the appropriate statistical model should reflect said multi-level variance structure. However, the cRSA approach only does so imperfectly with three major limitations, which we detail below.

First, the heterogeneity of subject- and/or condition-level variances in real datasets adversely impacts the sensitivity and reliability of cRSA. One of the fundamental assumptions of general linear models (step 4 of cRSA; see Figure 1D) is *homoscedasticity* or homogeneity of variance — that is, all residuals should have equal variance. This assumption is often violated in real datasets. The variance of first-level cRSA scores depends largely on the number of observations (see Figure 2), but this trial count can be highly variable across subjects or conditions, resulting in *heteroscedasticity*. This issue is prevalent in studies where trials are selected and grouped based on subject behavior, such as in memory or attention tasks. In those cases, treating cRSA scores obtained from 100 trials and those obtained from 20 trials as having equal variance would violate the homoscedasticity assumption and lead to unreliable results. One suggested remedy of the issue is equating the number of trials via random subsampling conditions with more trials (Dimsdale-Zucker & Ranganath, 2018); however, this solution comes with the cost of not making full use of the rich information from those low-variance conditions and may not be ideal when the number of trials is highly unbalanced.

Second, cRSA is unable to model stimulus-level variance. Many studies present subjects with a fixed set of stimuli, which are supposedly samples representative of some broader category. In this case, using cRSA brings about two issues. For one, stimuli can vary in a wealth of properties. Object images, for example, vary in terms of image complexity, concept frequency, and memorability, all of which can affect both behavior and neural activity in important and systematic ways (Bainbridge et al., 2017; Hovhannisyan et al., 2021; Naspi et al., 2021). To mitigate this issue, one might explicitly manipulate the distribution of relevant stimulus properties or cross-validate cRSA results on subsets of stimuli (Freund, Bugg, et al., 2021). Nevertheless, neither solution could address the second issue, which is the fact that the same stimuli are presented to multiple subjects. To properly address this stimulus-level dependence in the data and generalize the results beyond the fixed stimulus set, one must model stimulus identity as random effects, for the very same reason subjects are specified as random effects (Chen et al., 2021; Yarkoni, 2022). However, this solution is not accessible for cRSA since its current implementation stipulates the collapsing of stimulus information across trials (step 3 of cRSA; see Figure 1C). While other approaches such as pattern component modeling(Diedrichsen et al., 2018) and encoding models (Naselaris et al., 2011) are well-suited to analyzing variables that vary continuously on a trial-by-trial or moment-by-moment basis, these frameworks address different inferential goals. Specifically, pattern component modeling and encoding models focus on estimating variance components or predicting activation from features, while cRSA is designed to evaluate representational geometry. Thus, cRSA as well as our proposed approach address a problem setting distinct from pattern component modeling and encoding models.

Third, cRSA is not well-suited for testing the influence of stimulus-level or trial-level properties on neural representations. For example, it may be desirable to explicitly test or control for how certain stimulus-level properties (e.g., memorability) or trial-level recordings (e.g., cardiac cycle, pupil dilation) affect neural representations (Critchley & Garfinkel, 2018; van der Wel & van

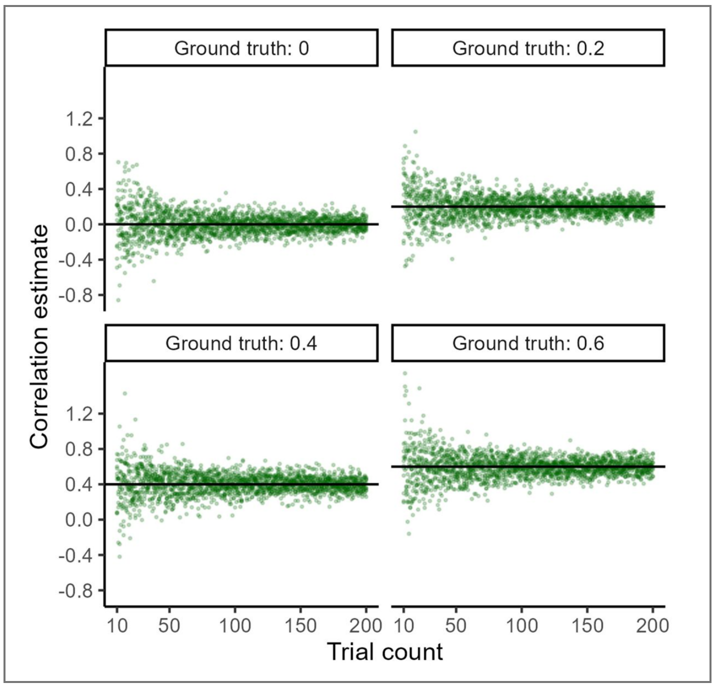


Figure 2. **Trial count influences the reliability of correlations.**

The spread of correlation estimates (y-axis) varies nonlinearly with the number of observations, or trial count (x-axis). Trial count ranges from 10 to 200 with increments of 1. For each trial count, observations were drawn from a bivariate normal distribution with the given ground truth Pearson correlation, and their empirical Pearson correlation coefficient was computed. Ten random samples were drawn for each trial count and ground truth combination. Both ground truth and estimated values in the scatter plots are Fisher-transformed (z).

Steenbergen, 2018). One strategy to analyze the effects of those stimulus- or triallevel continuous variables is to discretize them into categories (e.g., low vs. medium vs. high); however, discretization inevitably leads to a loss of information, reduced statistical power, and potentially misleading outcomes (Cohen, 1983; MacCallum et al., 2002). Alternatively, one could construct additional RSMs for the covariates and then compare all RSMs using a partial correlation or multiple regression framework, yet the statistical interpretations become much less straightforward after the conversion. For instance, even though a univariate random variable *v*, such as pleasantness ratings, can be conveniently converted to an RSM using pairwise distance metrics (Weaverdyck et al., 2020), the very same RSM would also be derived from the opposite random variable -v, leaving uncertain of the directionality (or if representation is strongest for pleasant or unpleasant items) of any findings with the RSM (see also Bainbridge & Rissman, 2018).

Here, we propose an original method for evaluating the neural representation of information at the level of individual experimental trials: *trial-level RSA* or *tRSA*. In this approach, instead of deriving a single measure of the similarity between $\text{RSM}_{\text{model}}$ and $\text{RSM}_{\text{brain}}$, we compute a series of similarity measures on a trial-by-trial (row-by-row or column-by-column) basis (Davis et al., 2021). For instance, the representational strength in the first trial is calculated as the similarity between the first row of $\text{RSM}_{\text{model}}$ and the first row of $\text{RSM}_{\text{brain}}$ (see Figure 1C). While several extensions of RSA have addressed key limitations in noise sensitivity, stimulus variance, and modeling (e.g., Diedrichsen et al., 2021; Schütt et al., 2023), our tRSA approach introduces a new methodological step by estimating representational strength at the trial level. This accounts for the multi-level variance structure in the data, affords generalizability beyond the fixed stimulus set, and allows one to test stimulus- or trial-level modulations of neural representations in a straightforward way. Critically, in single-presentation designs, a "trial" refers to one stimulus presentation, and corresponds to a row or column in the RSM. In studies with repeated stimuli, these rows are often called "conditions" and may reflect aggregated patterns across trials. tRSA is compatible with both cases: whether rows represent individual trials or averaged trials that create "conditions", tRSA estimates are computed at the row level. We assessed the efficacy of tRSA in comparison with cRSA using data from simulations representing a wide range of possible experimental scenarios (Experiment 1) and data from an extant real fMRI study (Experiment 2). In both experiments, our results demonstrated enhanced sensitivity, robustness, and flexibility with analyzing neural representations using the tRSA approach. Data is available upon request to the corresponding author and our simulations and example tRSA code is available at *https://github.com/electricdinolab*.

Methods

Experiment 1: cRSA and tRSA comparisons in simulated data

Experiment 1 consisted of three main parts. First, we validated the similarity between tRSA and cRSA results in assessing the general representational strength across all trials. In other words, we established that tRSA can carry out the main purpose of cRSA. Second, we compared the statistical inferences generated by each approach in a set of synthetic within-subject studies with two conditions. Third, we demonstrated tRSA's unique capability of assessing continuous modulators of representational strength in a set of synthetic scenarios. A host of parameters were varied to ensure the robustness of the comparisons in all three parts, such as the trial count, subject sample size, and effect size. All analyses were done in R 4.4.1 (R Core Team, 2024).

Estimating overall representational strength

We simulated *activity patterns*, from which RSMs were derived. Specifically, assuming an experiment with n trials and q measurement channels (e.g., voxels), we sampled a set of $n \cdot q$ values from a univariate standard normal distribution N(0,1), and another set of $n \cdot q$ values from a univariate normal distribution $N(0, \sigma^2)$. Both vectors were rearranged into matrices of n rows and q columns, yielding a *ground truth activity* pattern and a *noise* pattern. The summation of these two patterns yielded a *measured* pattern. Two RSMs were then generated from the ground truth pattern (RSM_{model}) and the measured pattern (RSM_{brain}) using Pearson's correlation, with which cRSA and tRSA were conducted separately. This set of procedures was repeated for 10,000 iterations with different random samples. The following parameter values were used for reported results: $q = 500, n \in \{10,15,20,25,30,40,50,100\}$, σ^2 ranging from 0.8 to 2.0 with increments of *0.2*.

We next assessed the correspondence between tRSA and cRSA in the presence of discrete conditions, which is often seen in real studies. Assuming a simple experimental design with two conditions, A and B, consisting of n_A and n_B trials, respectively, we generated a ground truth activity pattern with $n_A + n_B$ rows and m columns in the same way as before. Then, we generated two sets of noise patterns, which were controlled by parameters σ_A and σ_B, respectively, one for each condition. The measured pattern was again computed as the summation of ground truth and noise. cRSA was conducted separately for trials in each condition. Critically, tRSA was conducted in an *across-condition* fashion: we generated RSMs with all $n_A + n_B$ trials, obtained $n_A + n_B$ tRSA values as previously described, and then split the values into two sets based on the condition each trial belonged to. Finally, as before, we computed the average of tRSA values separately for Conditions A and B to allow comparisons with cRSA values. This set of procedures was repeated for 10,000 iterations with different random samples. We assessed the influence of three factors: sample size, balance, and noise level, with the following parameter values: $q = 500, n_A, n_B \in \{20,80,320\}$, $\sigma_A, \sigma_B \in \{1,2\}$.

Of note, tRSA could conceivably also be conducted in a *within-condition* fashion, whereby one would generate two separate sets of RSMs according to condition, i.e., one set of $n_A \times n_A$ RSMs for Condition A and another set of $n_B \times n_B$ RSMs for Condition B, and then compute the two sets of *within-condition* tRSA values separately. We hereby advocate for the use of *across-condition* tRSA — which we have used in previous studies (Davis et al., 2021; Howard et al., 2024; S. Huang, Howard, et al., 2024; Naspi et al., 2023) — over within-condition tRSA. Comparisons of the statistical properties of within-condition and across-condition tRSA approaches are discussed in Appendix 2.

Statistical inferences from tRSA and cRSA

Modeling discrete conditions

The essential theoretical advantage of tRSA over cRSA is that representational strength can be estimated at the level of experimental trials, which would allow us to properly capture the multilevel variance structure in the data. To demonstrate this benefit, four sources of variances were hypothesized: condition, subject, stimulus, and trial. Condition-level variance denotes the effect of experimental conditions or manipulations on representation, with each condition receiving its own noise-level parameter $s_{cond,k}$. This is the focal effect of interest for this simulation. Additionally, subject-level variance denotes individual differences in the quality of representation across m subjects, with each subject receiving one's own noise-level parameter $s_{subj,i}$. A within-subject design is common in real experiments and is thus assumed here, where each subject received all experimental conditions. Stimulus-level variance denotes the diversity of stimuli that may result in some being represented more strongly than others, with each stimulus receiving its own noise level parameter $s_{stim,j}$. The subject-level and stimulus-level variances were fully crossed, i.e., all subjects were assumed to view all stimuli exactly once (condition counterbalanced) during the experiment. Finally, trial-level variance denotes random fluctuations in the signal across the experiment and is controlled by a single noise-level parameter s_{trial}. Therefore, for a given event of subject i responding to stimulus j in condition k, we computed its noise level as:

$$\sigma_{ijk} = \max\left(S_{subj,i} + S_{stim,j} + S_{cond,k} + S_{trial}, 0.01\right).$$

Notably, this multi-level variance structure replaced the fixed noise parameters σ_A, σ_B, while all other procedures remained the same as the previous simulation. The entire set of procedures was repeated for 10,000 iterations. The base simulation parameters were as follows:

- Experimental design: $q = 500$, m = 40, $n_A = n_B = 100$, and
- Multi-level variance: $s_{cond} \in \{1,1.2\}$, $s_{subj} \sim N(0,0.5^2)$, $s_{stim} \sim N(0,0.2^2)$, $s_{trial} = 2$.

To assess the robustness of RSA methods in different situations that could occur in real data, we introduced a number of variations to the base design:

- Number of subjects: $m \in \{10,20,40,80,160,320\}$.
- Trial count per subject: $n_{trial} \in \{20,30,46,70,100,250,220,330,500\}$.
- Trial count ratio: $n_A \in \{20,40,60, \ldots,180\}, n_B = 200 — n_A$.
- Variance in trial count ratio across subjects: $\sigma_n \in \{0,1,2,4,8,16,32,64,128\}$. Actual trial counts were drawn from a truncated normal distribution, $n_A \sim truncN(100, \sigma_n, 10,190)$, $n_B = 200 — n_A$.
- Effect size: condition-level noise $s_{cond,A}/_B \in \{1.00,1.05,1.10,\ldots, 1.30\}$. Greater differences in the condition-level noise correspond to larger condition effects.

Following data generation, condition-level representational strength was estimated using cRSA and trial-level representational strength was estimated using tRSA. We focused on the statistical inferences of the effect of conditions, namely b_{B-A}. To this end, cRSA estimates were submitted to a paired-sample t-test. Critically, tRSA estimates were submitted to a mixed-effects model which is statistically appropriate for modeling the hierarchical structure of the data, where observations are nested within both subjects and stimuli (Baayen et al., 2008; Chen et al., 2021). Specifically, a linear mixed-effects model with a fixed effect of condition (which estimates the average effect across the entire sample, capturing the overall effect of interest) and random effects of both subjects and stimuli (which model variations due to differences between individual subjects and items, allowing generalization beyond the sample) were fitted to tRSA estimates via the 'lme4 1.1-35.3' package in R (Bates et al., 2015), and p-values were estimated using Satterthwaites's method via the 'lmerTest 3.1-3' package (Kuznetsova et al., 2017).

Given data generated with $s_{cond,A} = s_{cond,B}$, the correct inference should be a failure to reject the null hypothesis of $b_{B-A} = 0$; any significant ($p < 0.05$) result in either direction was considered a false positive (spurious effect, or Type I error). Given data generated with $s_{cond,A} > s_{cond,B}$, the inference was considered correct if it rejected the null hypothesis of $b_{B-A} = 0$ and yielded the expected sign of the estimated contrast ($b_{B-A} > 0$). A significant result with the reverse sign of the estimated contrast ($b_{B-A} < 0$) was considered a Type I error, and a nonsignificant ($p > 0.05$) result was considered a false negative (failure to detect a true effect, or Type II error). Error rates were computed for each RSA method and were compared against the null hypothesis of equal proportions between methods.

Modeling continuous modulators

In addition to discrete experimental conditions, we also simulated data reflecting scenarios in which representational strength varies continuously with some modulator. For example, one may be interested in the effects of continuous variables such as image complexity and memorability (stimulus-level) or reaction time (trial-level). In this case, s_{tri} was no longer a constant throughout the experiment but a trial-level variable. For a given event of subject i responding to stimulus j in trial k, we computed its noise level as:

$$\sigma_{ijk} = \max\left(S_{subj,i} + S_{stim,j} + S_{trial,k}, 0.01\right).$$

To capture the effect of this trial-level variable, we generated a random variable $v_{measured}$ that reflects a trial-level measurement with construct validity $r_{val} = Cor(V_{measured}, S_{trial})$, which was restricted to be a nonpositive value. When $r_{val} = 0$, $v_{measured}$ carries no information of the underlying s_{trial} and should not predict trial-level representational strength. When $r_{val} < 0$, $v_{measured}$ is inversely related to trial-level noise and should positively predict trial-level representational strength. The entire set of procedures was repeated for 10,000 iterations. The base simulation parameters were as follows:

- Experimental design: $q = 500, m = 40, n = 200$, and
- Variance structure: $r_{val} \in \{0,-0.1\}, S_{suj} \sim N(0,0.5^2), S_{stim} \sim N(0,0.2^2), s_{trial} = 2$.

As before, we introduced a number of variations, as follows:

- Number of subjects: $m \in \{10,20,40,80,160,320\}$.

- Number of trials per subject: $n_{trial} \in \{20,30,46,70,100,250,220,330,500\}$.
- Variance in the number of trials per subject: $\sigma_n \in \{0,20,40,\ldots, 160\}$. Actual trial counts were drawn from a truncated normal distribution, *n ~ truncN(n,* σ_n*, 20, ri).*
- Effect size: $r_{val} \in \{-0.10, -0.09,\ldots, 0.00\}$. More negative values of r_{val} should result in greater representational strength.

Following data generation, two conditions were created for each subject based on a median split (A=low $v_{measured}$, B=high $v_{measured}$), and we tested cRSA and tRSA models as described in the previous simulation, focusing on b_{B-A}. Furthermore, we additionally modeled trial-level representational strength as a function of $v_{measured}$ directly, using a linear mixedeffects model of tRSA estimates with a fixed effect of $v_{measured}$ and random effects of both subjects and stimuli. Given data generated with $r_{val} = 0$, the correct inference should be a failure to reject the null hypothesis of $b = 0$ in both discrete and continuous models; any significant ($p < 0.05$) result in either direction was considered a false positive or Type I error. Given data generated with $r_{val} < 0$, the inference was considered correct if it rejected the null hypothesis of $b = 0$ and yielded the expected sign of the estimated contrast or slope $b > 0$. A significant result with the reverse sign of the estimated contrast or slope ($b < 0$) was considered a Type I error, and a nonsignificant ($p >$ 0.05) result was considered a false negative or Type II error. Error rates were computed for each RSA method and were compared against the null hypothesis of equal proportions between methods.

Experiment 2: cRSA and tRSA comparisons in fMRI data

Experiment 2 proceeded in 3 steps. First, we assessed how the impact of subject-level variance in heterogeneity of trial ratios amongst two critical conditions ("Hits" and "Misses") can lead to biased estimates of representational strength. Second, we examined the impact of modeling trial-level variance on both representational strength during Object Naming, as well as Mnemonic Strength during a memory retrieval task. Lastly, stimulus-level estimates of representational strength were related to object memorability — an analysis that is not directly accessible to cRSA.

Data acquisition

Participants

A pre-existing dataset was analyzed to evaluate tRSA. Main study findings have been reported elsewhere (S. Huang et al., 2025). A total of 38 adults, aged 18 to 30, participated in this study on a voluntary basis and received monetary compensation for their time. Eligibility criteria required participants to be native or fluent English speakers, with no history of significant neurological or psychiatric conditions, and not taking medications that could affect cognitive function or cerebral blood flow (except for antihypertensive agents). All participants provided written informed consent before the start of the study. Six participants did not complete the study and were excluded from the analysis. Additionally, two participants were excluded from the Memory Retrieval analyses due to poor memory performance. The final sample included 32 participants (21 women and 11 men) for the Object Perception task and 30 participants (19 women and 11 men) for the Memory Retrieval task.

Experimental design

An expansive outline of the study design for the tasks described below can be found in (S. Huang et al., 2025); here we briefly review relevant details pertinent to our application of triallevel RSA for a subset of that data. Data from two experimental datasets were used, including an Object Perception dataset and a Memory dataset (Figures 7A and 8A). In the Object Perception task, participants were shown images of 114 unique everyday objects on a white background. The corresponding label presented underneath each object and participants were asked to rate how well the label described the object on a four-point scale. The main purposes of this rating task was to make sure that participants assess the meaning of the objects and to verify that these objects are indeed familiar to them (mean rating = 3.59). The objects were presented for 4 seconds, followed by a jittered fixation cross with an average duration of 4 seconds. In the memory task, a Memory Encoding session took place at least 7 days after Object Perception. During Encoding, participants viewed images of 114 unique real-world scenes along with the 114 objects they had seen in session 1. In each trial, participants were shown a scene image for 3 seconds, followed by a jittered empty box indicating the continuation of the trial, which lasted for an average of 3 seconds, and finally an object image for 4 seconds. During the object presentation, participants rated “how likely it is to find the object in the scene” on a 4-point scale (1 = “very unlikely,” 2 = “somewhat unlikely,” 3 = “somewhat likely,” 4 = “very likely”). Each trial was separated by a jittered fixation cross with an average duration of 4 seconds. Memory Retrieval consisted of three scanning runs, each with 38 trials, lasting approximately 9 minutes and 12 seconds. Memory Retrieval consisted of three scanning runs, each with 38 trials, lasting approximately 9 minutes and 12 seconds (within-run comparisons were later excluded from RSA analyses). Memory Retrieval took place one day after Memory Encoding and involved testing participants’ memory of the objects seen in the Encoding phase. Neural data during the Encoding phase has been reported elsewhere. In the main Memory Retrieval task, participants were presented with 144 labels of real-world objects, of which 114 were labels for previously seen objects and 30 were unrelated novel distractors. Participants performed old/new judgements, as well as their confidence in those judgements on a four-point scale (1 = Definitely New, 2 = Probably New, 3 = Probably Old, 4 = Definitely Old). In a subsequent Perceptual Memory Retrieval task participants were shown 126 images of real-world objects (96 old images, 18 were different exemplar images of old objects, and 12 images of unrelated novel objects). Participants were asked to determine whether the image was old, similar, or new. All analyses in the current study pertained to fMRI data from Object Perception and the main Memory Retrieval tasks, as well as behavioral data from the Perceptual Memory Retrieval task.

MRI data acquisition

MRI data were collected using a 3T GE MR750 Scanner equipped with an 8-channel head coil at the Brain Imaging and Analysis Center (BIAC) at Duke University. Each MRI session began with a localizer scan, during which 3-plane (straight axial/coronal/sagittal) faster spin echo images were obtained. A high-resolution T1-weighted (T1w) structural scan was then acquired, consisting of 96 axial slices parallel to the AC-PC plane, with voxel dimensions of $0.9 \times 0.9 \times 1.9$ mm^3. This was followed by blood-oxygenation-level-dependent (BOLD) functional scans using a whole-brain gradient-echo echo planar imaging sequence (repetition time = 2000 ms, echo time = 30 ms, field of view = 192 mm, 36 oblique slices with voxel dimensions of $3 \times 3 \times 3$ mm^3). Task instructions and stimuli were delivered using the PsychToolbox program (Kleiner et al., 2007) and projected onto a mirror at the back of the scanner bore. Participants responded using a four-button fiber-optic response box. To reduce scanner noise, participants wore earplugs, and MRI-compatible lenses were provided when necessary to correct vision. Foam padding was placed inside the head coil to minimize head movement.

Data analysis

MRI data preprocessing

fMRIPrep 23.0.1 (Esteban et al., 2019, 2023) was used to preprocess structural and functional MRI data, as well as generating text descriptions of preprocessing details, which were condensed below. T1w structural images collected across all MRI sessions for the same participant were corrected for intensity non-uniformity with ‘N4BiasFieldCorrection’ (Tustison et al., 2010) from ANTs 2.3.3 (Avants et al., 2011). The T1w-reference was skull-stripped with a Nipype implementation of the ‘antsBrainExtraction.sh’ workflow from ANTs, using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM), and gray-matter (GM) was performed on the brain-extracted T1w using ‘fast’ from FSL (Zhang et al., 2001). An anatomical T Iw-reference map was computed after registration of T 1w images using ‘mri_robust_template’ from FreeSurfer 7.3.2 (Reuter et al., 2010). Brain surfaces were reconstructed using ‘recon-all’ from FreeSurfer 7.3.2 (Dale et al., 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray matter of Mindboggle (Klein et al., 2017). Volume-based spatial normalization to the ICBM 152 Nonlinear Asymmetrical template version 2009c standard space was performed through nonlinear registration with 'antsRegistration' from ANTs 2.3.3, using brain-extracted versions of both T1w reference and the T1w template.

BOLD functional data across all sessions and runs were preprocessed collectively. A reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. Head-motion parameters with respect to the BOLD reference were estimated, followed by spatiotemporal filtering using 'mcflirt' from FSL (Jenkinson et al., 2002). BOLD runs were slice-time corrected to 0.972s (0.5 of slice acquisition range 0s-1.94s) using '3dTshift' from AFNI (Cox & Hyde, 1997). The BOLD time-series were resampled onto their original, native space by applying the transforms to correct for head-motion. The BOLD reference was then co-registered to the T1w reference using boundary-based registration via 'bbregister' from FreeSurfer (Greve & Fischl, 2009). Co-registration was configured with six degrees of freedom. Confounding time-series calculated based on the preprocessed BOLD included: root mean square displacement (RMSD) between frames (Jenkinson et al., 2002), absolute sum of relative framewise displacement (FD) (Power et al., 2014), and the derivative of root mean square variance over voxels (DVARS) (Power et al., 2014), as well as global signals extracted within CSF, WM, and the whole-brain mask. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor) (Behzadi et al., 2007). Principal components were estimated after high-pass filtering the preprocessed BOLD timeseries (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components were calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF+WM) were generated in anatomical space. For each CompCor decomposition, the k components with the largest singular values that cumulatively explained at least 50% of variance across the nuisance mask (CSF, WM, combined, or temporal) were retained. The BOLD time-series were resampled into standard space with a spatial resolution of $2 \times 2 \times 2$ mm^3 or $97 \times 115 \times 97$ voxels. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e., head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using 'antsApplyTransforms' (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Non-gridded (surface) resamplings were performed using 'mri_vol2surf' (FreeSurfer).

Single-trial modeling

Neural activity in gray matter (GM) voxels for each object presentation event was estimated using first-level least squares separate general linear models (Mumford et al., 2012) constructed with SPM12 (Friston et al., 2006) and custom MATLAB scripts. Subject-specific GM masks were generated by binarizing the fMRIPrep-derived probabilistic masks with a threshold set to exclude voxels with 80% probability of CSF or WM. Each model included a regressor for the event of interest along with a regressor for all other objects. Both regressors were convolved with the canonical double-Gamma hemodynamic response function, including their temporal and dispersion derivatives to account for variations in the timing of the peak response. Covariates of no interest included global signal, WM signal, CSF signal, FD, DVARS, RMSD, and six motion parameters related to translation and rotation. A high-pass temporal filter with a cutoff of 128 seconds was applied, and the AR(1) model was used to correct for autocorrelation. These first-level models yielded regression coefficients (betas) that estimated voxel-level neural activity corresponding to a single trial. Trials with FD greater than 1mm either before or during the trial were excluded (range: 0-28). Additionally, runs with more than 25% motion trials were excluded. In total, 1 Object Perception run, and 3 Memory Retrieval runs were excluded.

Behavioral Analyses

Trials for which participants gave no response in the Object Perception (mean 5) or the main Memory Retrieval (mean 4) were excluded from all analyses involving fMRI data at Perception and Memory Retrieval tasks respectively. As our primary behavioral measure, we assessed the memory performance for the object labels. Participants demonstrated variability in their ability to distinguish between old and new objects, as well as in their decision criteria, making direct comparisons of raw responses challenging. To address these biases, we performed a receiveroperating characteristics (ROC) analysis using 'yardstick 1.3.1' (Kuhn et al., 2024) in R. This analysis allowed us to categorize Hit and Miss trials by determining whether counting only "4" responses or both "3" and "4" responses as "old" yielded the best decision outcome (i.e., closest to 100% true positive and 0% false positive). The outcome of this analysis was then used to compute the sensitivity index (d') of recognition. Adjusted hit rates were used to determine adherence to the task and sufficient fMRI data during Memory Retrieval.

Behavioral analyses categorized each trial during Memory Retrieval as either a "Hit" or "Miss" for the purpose of categorizing the conditions in the fMRI data. This analysis determined whether counting only "4" responses or both "3" and "4" responses as "old" produced the most accurate decision outcome (i.e., closest to 100% true positive and 0% false positive). In total, data from 8 participants were adjusted to consider only "4" responses as "hit" trials. The results were then used to calculate the sensitivity index (d') of recognition within each subjective congruency condition and the corrected hit rate. Participants with post-adjustment hit rates below 40% were considered to have low adherence to the task and were excluded from all subsequent analyses. The number of included trials per participant for perception (mean = 110), ROC values (mean = 0.78), the count of hit trials (mean =76) as well as the corrected hit rate (mean = 0.70), are detailed in Appendix 3 Table 4.

Neural Representational Similarity Matrices

Models for neural pattern similarity (RSM_{brain}) were constructed using both the Object Perception study and the main Memory Retrieval task. Twenty six regions from the Human Brainnetome Atlas (Fan et al., 2016), focusing on eight areas within the lateral occipital cortex (LOC), 14 areas within the inferior temporal cortex (ITC), and four areas in the inferior parietal cortex (IPL, combining IPL sub regions into lateral posterior and anterior regions), as these regions are critically involved in visual representations during object perception and memory retrieval (Davis et al., 2021; Favila et al., 2020; Howard et al., 2024; Long & Kuhl, 2021). For each of these 26 regions, we constructed similarity matrices of voxel activation patterns across stimuli with custom scripts in MATLAB and SPM12 (Friston et al., 2006). This was done by vectorizing the voxel-level activation values within each region and calculating their correlations using Pearson's r, excluding all within-run comparisons. While each cell in a RSM_{model} reflects the similarity in stimulus properties, each cell in the 114×114 RSM_{brain} represents the similarity in activation patterns across different stimuli (Figure 1A).

Model Representational Similarity Matrix

The C2 layer of the Hierarchical Model of object recognition (HMAX) was used to capture visual similarity between the 114 objects. The PsyTorch implementation of the HMAX model was used and contains four sequential stages, each with their own output: S1, C1, S2, and C2. The S1 layer applies Gabor filters to the input image across multiple scales and orientations. The Gabor filters capture edge-like features such as bars and gratings. The output consists of feature maps that highlight these basic visual components. Following the S1 layer, the C1 layer performs local max pooling over the S1 feature maps. This operation introduces some degree of invariance to position and scale. By selecting the maximum response within localized regions, the C1 layer reduces the spatial resolution while retaining the most salient features. The stages S2 and C2 build upon this foundation using similar pooling mechanisms. Specifically, S2 units pool information from the C1 stage using linear filters and function as radial basis functions, responding most strongly to specific prototype input patterns. These prototypes are derived from random fragments extracted from a set of natural images, independent of the stimuli used in this study. The C2 layer then pools outputs from S2 units using a MAX operation, which provides a degree of position and scale tolerance, allowing for robust representation of visual objects. This global pooling results in a feature vector that is highly invariant to position and scale, capturing the presence of complex features regardless of their location in the input image. The C2 layer's output serves as a compact representation of the visual input (Kriegeskorte et al., 2008; Riesenhuber & Poggio, 1999, 2002; Serre et al., 2005, 2007; Sufikarimi & Mohammadi, 2020).

Features from the C2 layer of the HMAX model were used to create the model RSM. Pairwise similarity between images was computed using Pearson's correlation coefficient (r), measuring the relationship between their C2 feature vectors. These values were organized into a 114×114 RSM, where each cell represented the similarity between two images. The resulting RSM_{model} provided a structured representation of visual similarity, with higher correlations indicating shared complex features (Figure 1B).

Classic Representational Similarity Analysis (cRSA)

We conducted cRSA using data from both the Object Perception dataset and the Memory Retrieval dataset. For Object Perception, we estimated participants' classic representational strength in each of the 26 regions of interest by calculating Spearman's rho between the RSM_{brain} and the RSM_{model}, while excluding within-run comparisons and the diagonal cells of the matrices (see Figure 1C, Top). We then assessed the strength and reliability of visual representations across participants using one-sample t-tests (testing Fisher-transformed rho > 0) for each of the 26 regions, applying a false discovery rate (FDR) correction with a threshold of q = 0.05.

To evaluate cRSA's performance in detecting differences in representational strength between conditions, we analyzed data from the main Memory Retrieval task, focusing on old items. An RSM_{brain} for each of the 26 regions of interest was parsed into two separate RSMs: one containing data from "Hit" trials and the other from "Miss" trials. Similarly, the model RSMs were split based on participant responses. We then calculated Spearman's rhos for the Hit and Miss conditions by correlating RSM_{brain} and RSM_{model}, again excluding within-run comparisons and the diagonal cells of the matrices. These subject-specific representational strength measures were subsequently Fisher-transformed and analyzed using paired-sample t-tests. Regions showing significantly higher representational strength for Hit trials compared to Miss trials, after FDR correction with q < 0.05, were identified as *mnemonic representation regions*.

To assess the impact of unbalanced trial counts on mnemonic representational strength, we calculated a Contrast Variance Factor (CVF) for each participant in regions showing evidence of mnemonic representation ($p < 0.05$, uncorrected). The participant contrast score was determined by subtracting the Miss representational strength ($R_{i,M}$) from the Hit representational strength ($R_{i,H}$).

$$D_i = R_{i.H} - R_{i.M}$$

Next, we computed the mean difference $\overline{D}$ across all participants:

$$\overline{D} = \frac{1}{N}\sum_{i}^{N} D_i$$

To quantify how much each participant's mnemonic representational strength in each ROI deviated from the mean (thus inflating the contrast variance), we calculated the normalized absolute difference between the participant contrast score (D_i) and the average contrast score ($\overline{D}$):

$$CVF_i = \frac{\left|D_i - \overline{D}\right|}{\overline{D}}$$

Finally, to evaluate the influence of unbalanced trial counts on the variance factor, we correlated the participant CVF averaged across regions with the ratio of Hit trial count ($N_{i,H}$) to Miss trial count ($N_{i,M}$).

$$Ratio_i = \frac{N_{i.H}}{N_{i.M}}$$

Trial-Level Representational Similarity Analysis (tRSA)

The tRSA approach used the same model and neural RSMs from the Object Perception dataset and the main Memory Retrieval dataset as described above. However, instead of correlating the two matrices in their entirety, correlations were computed on a row-by-row basis, excluding within-run similarities and the diagonal cells of the matrices (see Figure 1C, Bottom). The triallevel estimates from both phases were then fitted to a series of mixed-effects models using the 'Ime4' package in R (R Core Team, 2024) with restricted maximum likelihood methods. These fitted models were then evaluated using the Akaike Information Criterion (AIC) to determine the optimal random effect structure (Meteyard & Davies, 2020; Park et al., 2020). To further validate model selection (Matuschek et al., 2017), models were refit using maximum likelihood estimation and subjected to model selection via log-likelihood ratio tests (LRTs). The average AICs and the results of the LRTs are reported in Appendix 3 Table 5. Restricted maximum likelihood models with the selected random effects structure were further analyzed using the 'ImerTest' package. Denominator degrees of freedom were estimated using Satterthwaite's method (Satterthwaite, 1946), and fixed effects were tested using t-tests with an alpha level of 0.05, corrected for false discovery rate (FDR) to account for multiple comparisons.

To estimate tRSA representational strength, trial-level estimates from the Object Perception dataset were fit to a mixed-effects model with random intercepts for Participant and Stimulus. Estimated intercepts ($b > 0$) were used to identify representational regions. For detecting tRSA mnemonic representations in the main Memory Retrieval task, instead of constructing separate Hit and Miss RSMs, row-wise correlations were performed across all trials in the matrices (i.e., across-condition tRSA). This approach allows for trial-level estimates of representational strength that can be subsequently categorized as "Hit" or "Miss" and analyzed using a series of mixed-effects models. The model for mnemonic representation data included the fixed effect of Memory Success (Hit vs. Miss trials) and random intercepts for Participant and Stimulus.

Examining continuous modulators of representation

The capacity of tRSA in examining stimulus-level variance modulating representation was assessed. For each item, a memorability score was calculated as the average response across participants (on a scale of 1-4 for the main Memory Retrieval and 1-3 for Perceptual Retrieval), normalized by the maximum possible value (4 for main retrieval and 3 for perceptual retrieval). This approach produced two stimulus-specific continuous variables indicating the overall confidence with which each item would be recalled. We refer to these measures as Conceptual Memorability and Item Memorability (perceptual). The Item Memorability measure was used as an of-interest fixed effect for trial-level representational strength estimates from the Object Perception dataset. Additional fixed nuisance variables included Conceptual Memorability, fMRI run, and trial-level reaction time. Random intercepts for Participant and Stimulus were also included.

Results

Experiment 1: cRSA and tRSA comparisons in simulated data

Basic statistical properties of tRSA estimates

We simulated activity patterns from which RSMs were then derived and used to estimate representational strength. Notably, while it is difficult to fix the ground truth representational strength, we manipulated the statistical dependence between the activity patterns — and by extension, between RSM_{brain} and RSM_{model} — by injecting varying levels of measurement noise (σ^2). Simulations indicated that increasing measurement noise indeed reduced estimates of representational strength by both cRSA and tRSA (see Figure 3A). Also as expected, greater trial counts led to more stable estimates for both methods. Most importantly, a strong positive correlation between cRSA and tRSA estimates can be observed even in the “noisiest” simulation (i.e., $n = 10, \sigma^2 = 2$; Intercept $b_0 = 0.00$, SE = 0.01, t = 0.33, p = 0.74; Slope $b_{cRSA} = 1.00$, SE = 0.02, t = 64.95, p < 0.001). These results demonstrated the close numerical correspondence between tRSA and cRSA in estimating the overall representational strength of a given system across all trials.

Discrete conditions

Oftentimes, researchers are interested in not only representational strength *per se*, but also how representation *changes* with other factors like experimental manipulations (e.g., “easy” vs. “hard”), behavioral performance (e.g., “remembered” vs. “forgotten”), or some combination. Analyzing these differences with cRSA entails partitioning the data to generate separate condition-specific RSMs. Subsequently, an intuitive version of *within-condition* tRSA can be computed using those RSMs, and its estimates would closely track cRSA estimates, as we have demonstrated in the previous section. Instead, we advocate for *across-condition* tRSA: full RSMs are used for computing trial-level representational strength, and the estimates are then split by conditions as necessary. For a direct comparison of *within-* and *across-condition* tRSA, see Appendix 2.

Here, we examined the correspondence of *across-condition* tRSA with cRSA in a wide range of scenarios where data can be meaningfully split into two subsets. Specifically, we assessed the influence of three important factors that may vary widely across studies: *raw trial counts*, *balance of trial counts*, and *effect sizes*. Because cRSA was performed separately for each condition, changes in the trial count or effect size of Condition B had no influence on cRSA estimates for Condition A. However, they indeed affected estimates obtained from *across-condition* tRSA, where all experimental trials were used for representational strength estimation. Specifically, all else being equal, increasing the trial count in Condition B from 20 to 80 or from 80 to 320 improved the reliability of tRSA estimates in Condition A (see Figure 3B, Rows 1 and 3). Additionally, increasing the noise level for Condition A to be higher than that for Condition B led to reduced tRSA estimates in Condition B as well. These effects were also reflected in the condition difference in tRSA estimates (see Figure 3C). Moreover, it is critical to note that tRSA appeared to be more reliable than cRSA in almost all simulated scenarios, especially in terms of the estimated difference between conditions. These findings suggest that tRSA may be more sensitive to true effects and at lower risk of false positives, i.e., reduced Type II and Type I error rates in formal statistical tests.

Contrasting conditions

The next set of simulations compared across-condition tRSA to cRSA in terms of statistical inferences, focusing on the difference in representational strength between two conditions as a typical effect of interest in many empirical studies. We implemented the condition differences by altering two condition-level noise parameters, where more noise would result in lower representational strength. The statistical significance of the condition difference b_{B-A} was estimated by a paired-sample t-test for cRSA and by a linear mixed-effects model for tRSA, using a threshold of $\alpha = 0.05$. Depending on whether the two conditions ought to have the same quality of representation, outcomes were classified as true positives, true negatives, false positives (Type I error), or false negatives (Type II error). The performance of cRSA and tRSA were quantified with their specificity (better avoids false positives, 1 - Type I error rate) and sensitivity (better avoids false negatives 1 - Type II error rate).

We first examined how cRSA and tRSA performed with effects of different sizes, under a base within-subject experimental design consisting of 40 subjects and 200 unique stimuli randomly and evenly split between two conditions. Across 10,000 simulations, cRSA and tRSA were comparable in terms of specificity, with both Type I error rates neighboring around the nominal 5% (see Figure 4, diagonal cells in red). As the true condition difference increased, the sensitivity of both methods also increased accordingly. Importantly, tRSA were significantly more sensitive to true effects than cRSA in a range of simulated scenarios, consistent with our hypothesis (see Figure 4, off-diagonal cells in blue; asterisks indicate where tRSA was statistically more sensitive then cRSA).

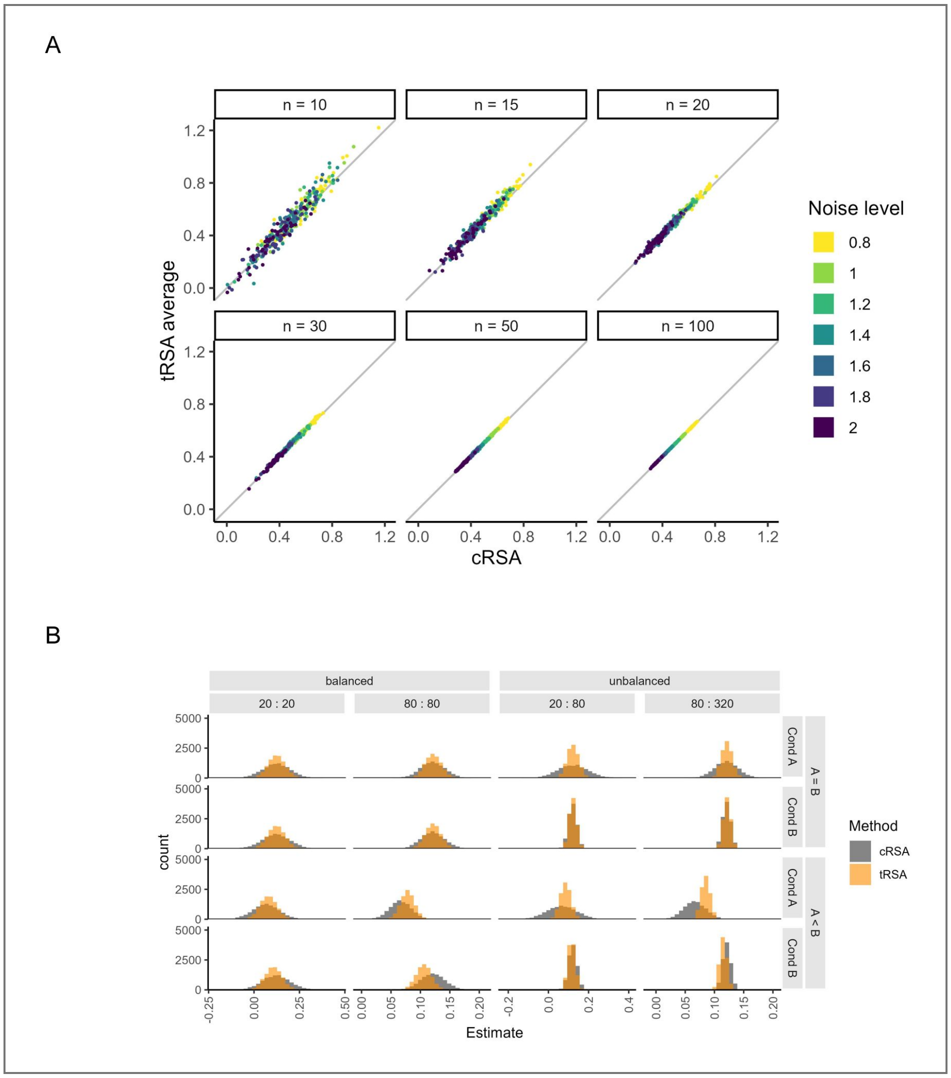


Figure 3. **Correspondence between classic and trial-level RSA.**

A) Scatter plots of overall representational strength estimates, based on simulated activity patterns, produced by cRSA (x-axis) and tRSA (y-axis). Trial count (n) and measurement noise level ($\sigma^{\%}$) were both varied. 50 iterations were performed for each parameter combination. Solid slopes indicate where tRSA equaled cRSA (y = x). **B)** Histograms of overall representational strength estimates, based on simulated activity patterns for two separate conditions, produced by cRSA (gray) and across-condition tRSA (orange). Each histogram depicted 10,000 iterations. Trial counts were varied; for example, "20 : 80" means that Condition A contained 20 trials and Condition B contained 80 trials, which was an unbalanced scenario. Effect sizes were also varied, such that the ground truth representational strength would be equal between conditions ("A = B") or stronger in Condition B ("A < B"). **C)** Histograms of condition differences in representational strength values in **B**.

Figure 3. (continued)

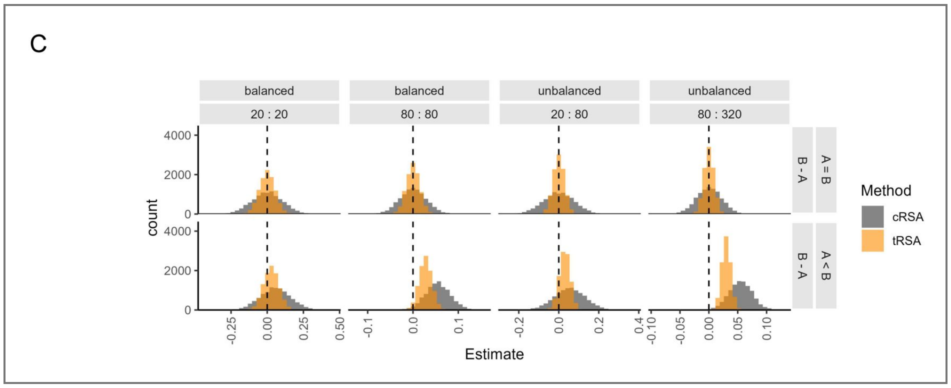
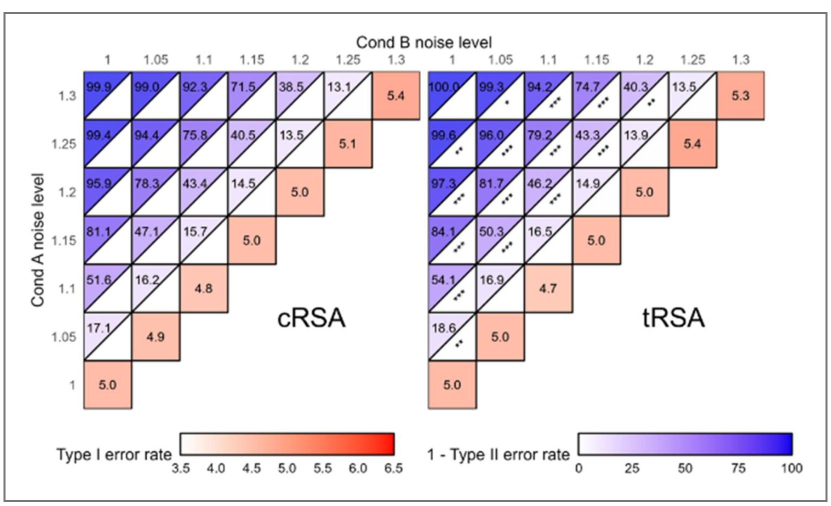


Figure 4. Testing condition differences in representational strength with varying effect sizes.

The effect size of the condition difference in representation was manipulated by changing the noise level in each condition in small increments, with higher noise levels corresponding to lower ground truth representational strengths. When the noise level was the same for both conditions, Type I error rates (red) were computed as the proportion of significant contrasts across 10,000 iterations, regardless of sign of the estimate. Otherwise, proportions were computed separately for effects of the correct sign (+, or B>A; blue) and of the incorrect sign (-, or B<A). Asterisks indicate the significance level of the test of equal proportions between simulation results from cRSA (left) and tRSA (right). Significance annotation: *** p < 0.001, ** p < 0.01, * p < 0.05.



We further probed the robustness of cRSA and tRSA across datasets that varied in several important experimental design factors, namely the number of participants, trial counts per participant, and the balance of trial counts. In terms of specificity, our simulations did not suggest an effect of those manipulated factors, as the proportions of significant contrasts yielded by both cRSA and tRSA did not significantly deviate from the nominal alpha level of 5%. Tests of significant contrast proportions reported by cRSA and by tRSA also failed to reject the null hypothesis of equal proportions ($p > 0.05$), suggesting similarly acceptable Type I error rates of both approaches (see Figure 5, Left).

In terms of sensitivity, our simulations revealed large impacts of all manipulated parameters. Increasing the number of subjects or the trial count per subject improved the sensitivity of true effects, benefiting both cRSA and tRSA substantially. Tests comparing the sensitivity of tRSA and cRSA suggested an advantage of tRSA at small-to-medium sample sizes of up to at least 40 subjects, and at medium trial counts of between 100 and 220 trials per subject (see Figure 5A-B, Right). We additionally tested variabilities in trial counts that may occur in real experiments. In one set of simulations, the total trial count was fixed at 200 but the ratio of trial counts between conditions varied consistently across subjects (i.e., systematic imbalance; Figure 5C). In another set of simulations, while trial counts were kept approximately equal between conditions *on average*, subject-specific ratios varied (Figure 5D). In both scenarios, increasing heterogeneity in trial counts resulted in substantially reduced sensitivity for cRSA. This result was expected: the reliability of cRSA estimates depends largely on trial counts (Figure 2), which violated the homoscedasticity assumption of the subsequent paired-sample t-test. In the meantime, with trial-level estimates of representational strength from tRSA, the subsequent mixed-effects models were able to properly handle this variance in trial count. Indeed, our simulations validated that tRSA was more robust to extreme scenarios and always significantly outperformed cRSA in sensitivity (see Figure 5C-D, Right).

Continuously varying effects

A crucial advantage of tRSA over cRSA is that this approach offers a straightforward way to model representational strength as a function of stimulus- or trial-level measures, such as image complexity and subjective familiarity. While these measures could be discretized into a much smaller number of bins (e.g., “low” and “high”), they are better treated as continuous variables to preserve meaningful variance. To simulate these scenarios, we simulated a trial-level measurement variable $v_{measured}$ that hypothetically tracks trial-level variability s_{trial} with construct validity r_{val} = Cor $v_{measured}$, S_{trial}). With higher construct validity, $V_{measured}$ becomes more predictive of trial-level variability in noisiness and, therefore, representational strength. To perform cRSA, trials were post-hoc split into two conditions by performing a median split on $v_{measured}$ for each participant. Statistical inferences were again drawn for the difference in representational strength between the two conditions, using both cRSA and tRSA$_{\text{discrete}}$. In addition, we also assessed the performance of tRSA$_{\text{continuous}}$, where representational strength was directly predicted by the measured continuous variable $v_{measured}$ without discretization.

Specificity and sensitivity were first examined for datasets with various effect sizes. Across 10,000 iterations, cRSA had a Type I error rate (i.e., when $r_{val} = 0$) close to the nominal rate of 5%, while both tRSA$_{\text{discrete}}$ and tRSA$_{\text{continuous}}$ showed significantly improved specificity (see Figure 6A). As construct validity and effect size of the trial-level measurement increased (i.e., more negative r_{val}), sensitivity expectedly improved for all methods. Relative to cRSA, tRSAcontinuous appeared to show slightly lower sensitivity for small effects (e.g., $r_{val} = -0.01$), though it was also more conservative than cRSA with a lower chance of declaring significance in the incorrect direction (B<A). Moreover, tRSA$_{\text{continuous}}$ was the most sensitive method for moderate effects (e.g., $r_{val} =$ -0.04). We further probed the robustness of cRSA and tRSA across datasets that varied widely in the number of subjects and trial counts. Overall, we observed that the Type I error rates were stable around the nominal 5% for cRSA, as expected. However, both tRSA$_{\text{discrete}}$ and tRSA$_{\text{continuous}}$ demonstrated significantly enhanced specificity across all manipulations (see Figure 6B-D, Left). Sensitivity expectedly improved for all three approaches with increasing subject numbers and trial counts and hit the ceiling with enough samples, though tRSA$_{\text{continuous}}$ significantly

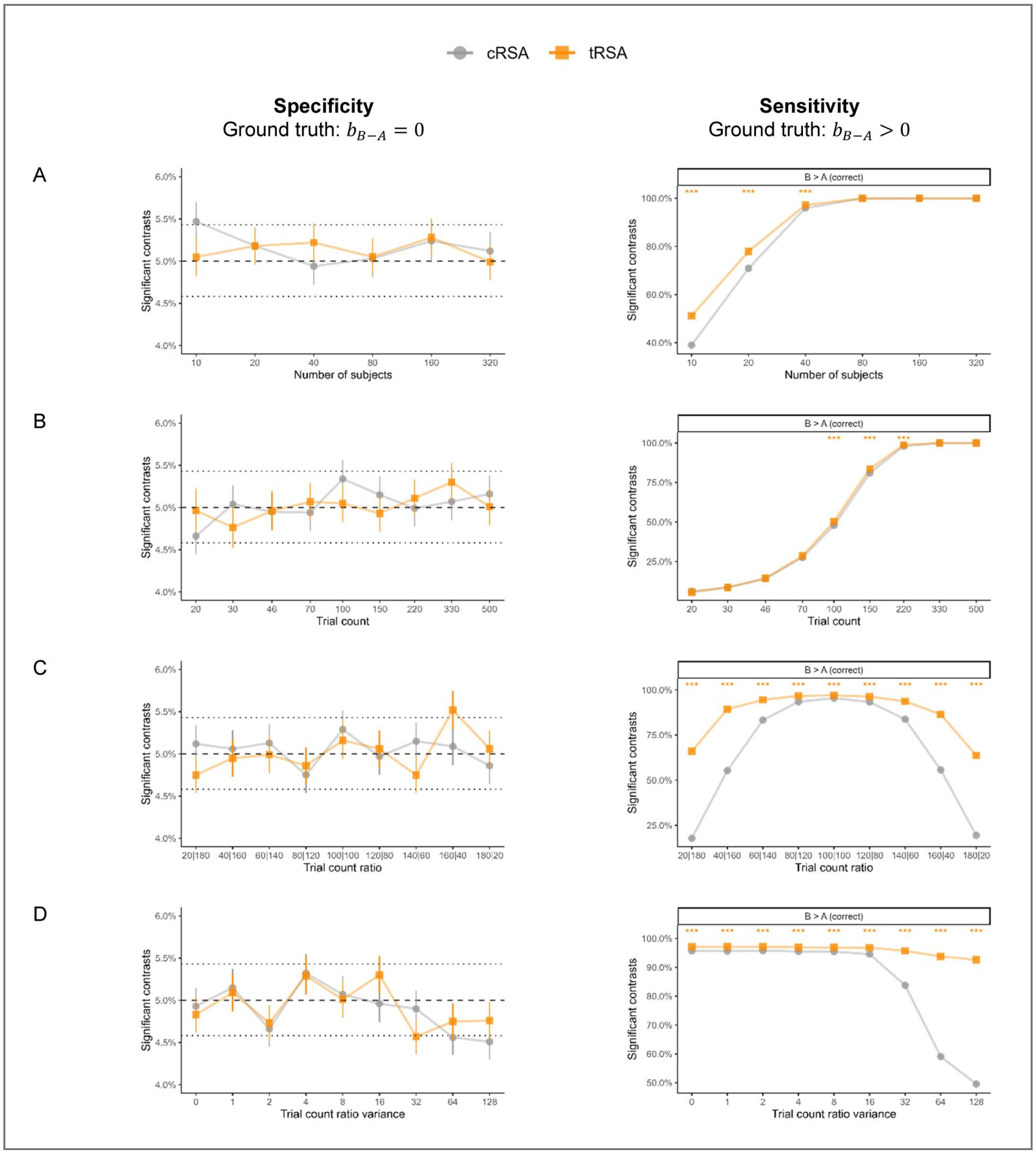


Figure 5. **Testing condition differences in representational strength with varying designs.**

The robustness of cRSA and tRSA were assessed in several variations of a base experimental design consisting of 40 subjects and 200 unique stimuli (randomly and evenly split into two sets). Simulations varied in **A)** number of subjects, **B)** trial count per subject, **C)** the ratio of trial counts between two conditions (with a fixed total), and **D)** the variance of trial count ratios across participants (even split overall). In each simulation, the statistical significance of the condition difference in representational strength was determined by a paired-sample t-test for cRSA and a linear mixed-effects model for tRSA, with α = 0.05. The proportion of significant contrasts was computed across 10,000 iterations. Type I error rates were computed regardless of the direction of the contrast. Type II error rates were computed only for the correct contrast (i.e., B > A). Error bars indicate standard errors. In the left column, dashed horizontal lines mark the nominal α level of 0.05, and dotted horizontal lines mark the critical values beyond which the estimated proportion would be significantly different from α. Asterisks indicate the significance of the deviation in error rates between cRSA and tRSA results, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

outperformed other methods at smaller sample sizes, such as 10 participants with 200 trials each or 40 participants with 30 trials each (see Figure 6B-C, Right). Additionally, we simulated scenarios in which subject-level trial counts were variable (e.g., missingness, exclusions), resulting in not only a reduction of total sample size but also additional variance in any subject-level estimates. The sensitivity of cRSA dropped substantially with this change because cRSA does not account for such variance, while tRSA-based inferences were minimally impacted (see Figure 6D, Right).

Summary of Experiment 1

Collectively, our simulations offered three key insights into the commonalities and distinctions between our novel tRSA and the commonly implemented cRSA approaches. First, when the goal is to quantify the overall representational strength of the entire set of trials, tRSA produces estimates highly similar to those generated by cRSA. In other words, little information is lost in performing the trial-level computations (Figure 3). Second, tRSA is robust to variances that cRSA is agnostic to, such as imbalances in trial count between conditions and variability in trial count ratios across subjects. In almost all cases tRSA significantly outperformed cRSA in sensitivity (Figure 5). This outcome was expected, as the collapsing of trials in cRSA inevitably leads to the loss of important information regarding meaningful variances from other levels; tRSA is capable of properly modelling the multi-level variance structure. Third, tRSA is uniquely advantageous when neural representations are to be linked to continuous variables, rather than discrete conditions. The voided need for post-hoc discretization prevents additional loss of meaningful variance from the modulators, and $\text{tRSA}_{\text{continuous}}$ demonstrated significantly enhanced specificity and sensitivity over cRSA in a wide range of scenarios (Figure 6).

Experiment 2: cRSA and tRSA comparison in fMRI data

Trial-level RSA estimates for Perceptual and Mnemonic Processing

The first goal of Experiment 2 was to compare the results from tRSA and cRSA approaches obtained from real fMRI data. We used fMRI data collected when participants implicitly named 114 colored images of everyday objects in a white background (Object Perception; see Figure 7A). Estimates of regional representational strength in the tRSA models were compared with mean cRSA values across participants, for each region of interest. Unsurprisingly, the tRSA model estimates and the cRSA mean correlation values demonstrated similar distributions (Figure 7B) and were highly correlated (Figure 7C). Critically, however, the two approaches yielded different results in terms of statistically significant regions. Specifically, the cRSA approach identified four significant regions in the LOC during Object Perception, whereas the tRSA approach identified two additional regions within the LOC (Table 1, Figure 7D, and Figure 7E). This discrepancy indicated the ability of tRSA to more accurately and robustly identify representational regions.

Subject-level variance not accounted for by cRSA

The second goal of Experiment 2 was to demonstrate the utility of tRSA in accounting for the heterogeneity in subject-level variance. In the main Memory Retrieval task, experimental trials were categorized as “Hit” or “Miss” based on participants’ memory success, and the exact number of trials in each condition varied widely across participants (Figure 8A, Bottom). Importantly, the difference in the number of trials had a strong impact on the estimation of representational strength (see Figure 2). Participants with fewer trials in a given condition exhibited a noticeably wider distribution of representational strength values across ROIs in that condition, indicating that trial count was a substantial source of across-subject variance indeed (see Figure 8B). Moreover, trial count was also negatively associated with the standard deviation of representational strengths across ROIs, for both Hit trials (rho = -0.37, $p = 0.043$) and Miss trials (rho = -0.71, $p < 0.001$). However, this subject-level variance is often not accounted for in subsequent group-level t-test or ANOVA, which assumes that each participant contributes equally reliable estimates (i.e., homoscedasticity). Indeed, CVFs were much higher for participants with more extreme Hit to Miss trial counts (rho = 0.45, $p = 0.016$; see Figure 8C).

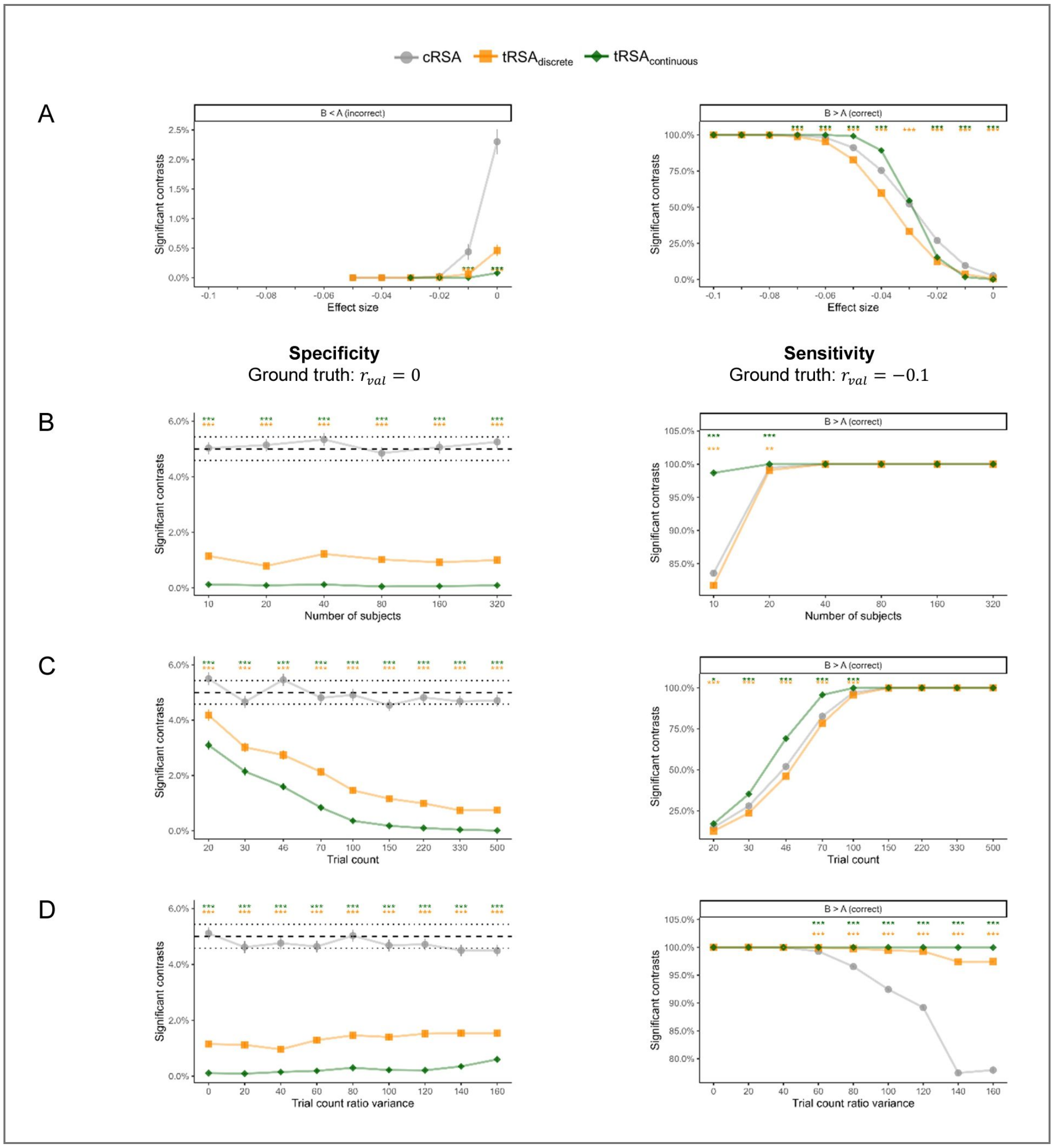


Figure 6. Testing continuous modulations of representational strength with varying designs.

The robustness of cRSA and tRSA were assessed in several variations of a base experimental design consisting of 40 subjects and 200 unique stimuli presented once for each subject. Simulations varied in **A)** the true effect size of representational strength, **B)** number of subjects, **C)** trial count per subject, and **D)** the variance of trial count across subjects (assuming a fixed total). Three statistical tests were performed in each simulation: a paired-sample t-test on the difference between "high" and "low" conditions (median-split of $v_{measured}$) for cRSA, a linear mixed-effects model on the same condition difference for $tRSA_{discrete}$, and a linear mixed-effects model on the effect of continuous variable $v_{measured}$ for $tRSA_{continuous}$. Statistical significance was determined with α = 0.05. The proportion of significant results was computed across 10,000 iterations. Type I error rates were computed regardless of the sign of the estimate. Type II error rates were computed only for the correct sign (+). Error bars indicate standard errors. In the left column for B through D, dashed horizontal lines mark the nominal α level of 0.05, and dotted horizontal lines mark the critical values beyond which the estimated proportion would be significantly different from α. Asterisks indicate the significance level of the test of equal proportions between each tRSA approach and cRSA, *** p < 0.001, ** p < 0.01, * p < 0.05.

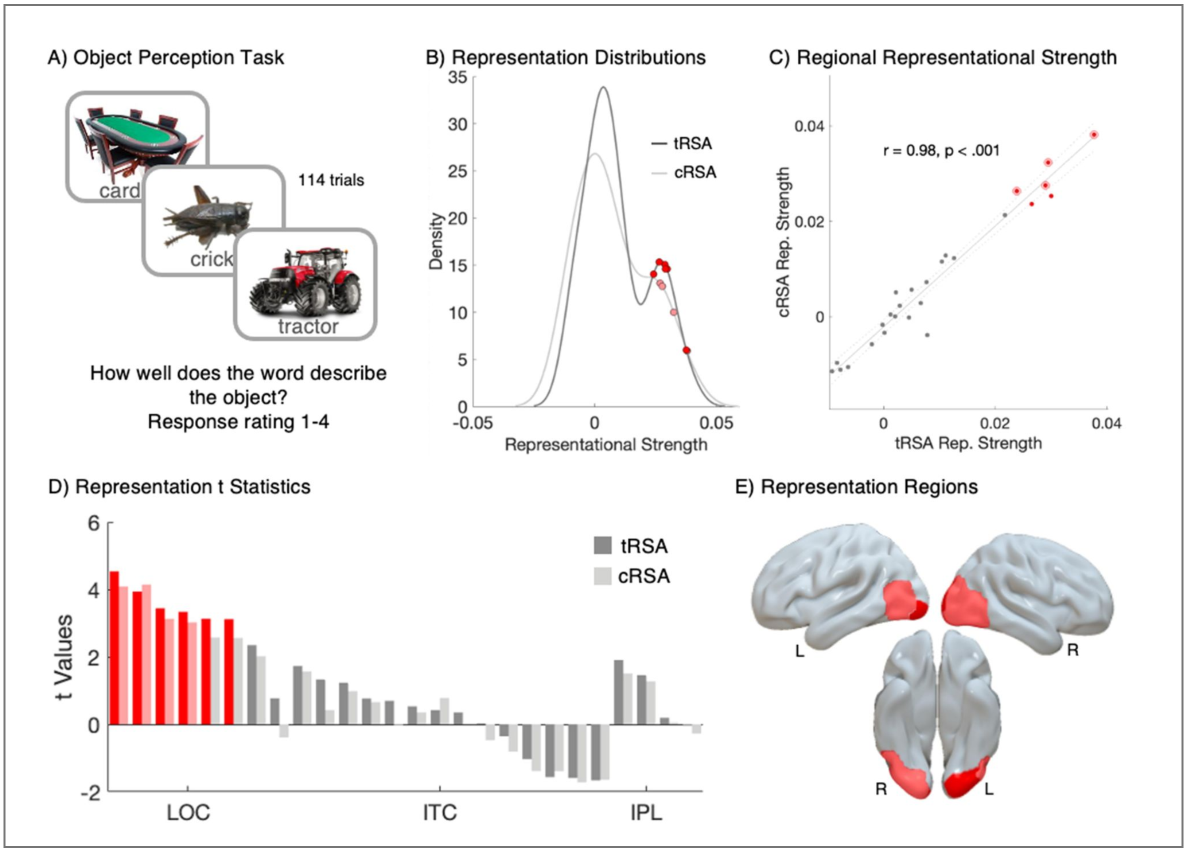


Figure 7. **tRSA outperforms cRSA in identifying representational regions during object perception.**

A) Visualization of the Object Perception Task. The fMRI task required participants to view 114 labeled objects and rate how well the labels describe the objects on a scale of 1 to 4 (mean 3.59). **B)** Density plots (y-axis) of regional representational strengths (x-axis) for tRSA (model estimates, dark) and cRSA (mean correlation values, light). Regions with significant representational strength greater than zero ($q < 0.05$) are highlighted in red. **C)** Scatter plot showing the correlation between regional representational strengths from tRSA (model estimates, y-axis) and cRSA (mean correlation values, x-axis). Regions with significant tRSA representation are highlighted in red, and those significant in cRSA are outlined in red. The representational strength measures are highly correlated across regions ($r = 0.98$, $p < 0.001$). **D)** Bar graph of t-statistics for each of the 26 regions of interest. Representation t-statistics (y-axis) from cRSA (light) and tRSA (dark) across 26 regions of interest (x-axis). Regions significant after FDR correction ($q < 0.05$) are highlighted in red. E) Brain regions with significant representation. Regions in light red were identified by the tRSA and cRSA. Regions in dark red were identified by the tRSA method only.

			cRSA					tRSA				
REGION	SubArea	Coordinates (x, y, z)	Mean	SE	t Values	p	q	coeff	SE	t Values	p	q
LEFT IPL	A40c, caudal area 40(PFm)	-56, -49, 38	-0.002	0.006	-0.276	0.785	0.887	0.000	0.005	-0.039	0.969	0.980
RIGHT IPL	A40c, caudal area 40(PFm)	57, -44, 38	0.000	0.008	0.053	0.958	0.992	0.001	0.006	0.199	0.843	0.914
LEFT IPL	A39rv, rostroventral area 39(PGa)	-47, -65, 26	0.012	0.009	1.273	0.212	0.394	0.010	0.007	1.461	0.154	0.307

Table 1. **Representation regions identified by cRSA and tRSA.**

RIGHT IPL	A39rv, rostroventral area 39(PGa)	53, -54, 25	0.012	0.008	1.511	0.141	0.333	0.013	0.007	1.901	0.066	0.215
LEFT ITC	A20iv, intermediate ventral area 20	-45, -26, -27	0.002	0.007	0.352	0.727	0.859	0.003	0.005	0.533	0.598	0.777
LEFT ITC	A20r, rostral area 20	-43, -2, -41	-0.011	0.007	-1.383	0.178	0.357	-0.006	0.006	-1.027	0.314	0.510
LEFT ITC	A20il, intermediate lateral area 20	-56, -16, -28	-0.011	0.006	-1.722	0.095	0.309	-0.008	0.005	-1.587	0.123	0.278
RIGHT ITC	A20iv, intermediate ventral area 20	46, -14, -33	-0.010	0.007	-1.385	0.176	0.357	-0.008	0.005	-1.563	0.128	0.278
RIGHT ITC	A20r, rostral area 20	40, 0, -43	0.005	0.006	0.781	0.442	0.676	0.002	0.005	0.419	0.679	0.824
RIGHT ITC	A20il, intermediate lateral area 20	55, -11, -32	-0.011	0.007	-1.642	0.111	0.320	-0.009	0.006	-1.660	0.107	0.277
LEFT ITC	A37elv, extreme lateroventral area37	-51, -57, -15	-0.003	0.007	-0.470	0.642	0.859	0.000	0.006	0.025	0.980	0.980
LEFT ITC	A37vl, ventrolateral area 37	-55, -60, -6	0.003	0.007	0.418	0.679	0.859	0.007	0.005	1.331	0.193	0.358
LEFT ITC	A20cl, caudolateral of area 20	-59, -42, -16	0.013	0.008	1.569	0.127	0.329	0.011	0.006	1.728	0.094	0.271
LEFT ITC	A20cv, caudoventral of area 20	-55, -31, -27	0.006	0.009	0.657	0.516	0.746	0.005	0.007	0.765	0.450	0.651
RIGHT ITC	A37elv, extreme lateroventral area37	53, -52, -18	-0.006	0.007	-0.807	0.426	0.676	-0.002	0.006	-0.354	0.726	0.824
RIGHT ITC	A37vl, ventrolateral area 37	54, -57, -8	0.000	0.007	0.010	0.992	0.992	0.002	0.006	0.350	0.729	0.824
RIGHT ITC	A20cl, caudolateral of area 20	61, -40, -17	0.000	0.008	-0.019	0.985	0.992	0.005	0.007	0.696	0.492	0.673
RIGHT ITC	A20cv, caudoventral of area 20	54, -31, -26	0.007	0.007	0.983	0.333	0.577	0.008	0.006	1.233	0.227	0.393
LEFT LOC	mOccG, middle occipital gyrus	-31, -89, 11	0.021	0.011	2.015	0.053	0.196	0.022	0.009	2.345	0.023	0.086
LEFT LOC	V5/MT+, area V5/MT+	-46, -74, 3	0.032	0.010	3.126	0.004	0.033	**0.030**	**0.009**	**3.440**	**0.002**	**0.010**
LEFT LOC	OPC, occipital polar cortex	-18, -99, 2	-0.004	0.010	-0.393	0.697	0.859	0.008	0.010	0.772	0.442	0.651
LEFT LOC	iOccG, inferior occipital gyrus	-30, -88, -12	0.024	0.009	2.560	0.016	0.067	**0.027**	**0.009**	**3.115**	**0.003**	**0.013**
RIGHT LOC	mOccG, middle occipital gyrus	34, -86, 11	0.038	0.009	4.082	0.000	0.004	**0.038**	**0.008**	**4.533**	**0.000**	**0.001**

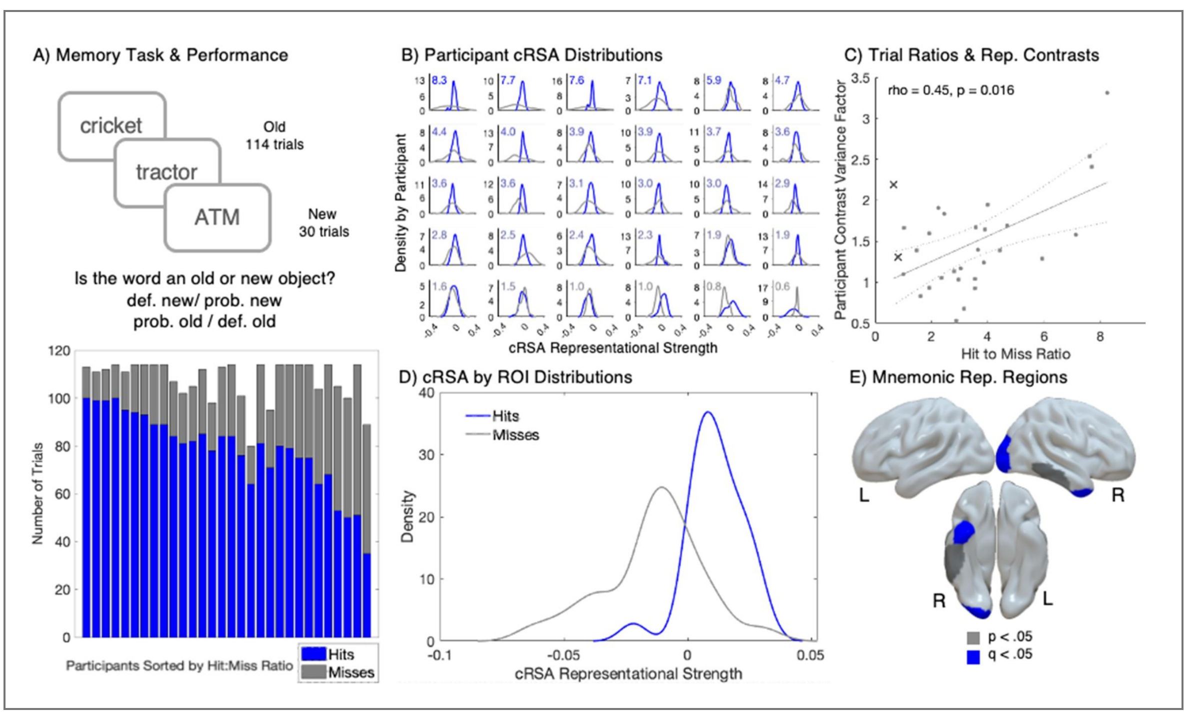
Table 1. (continued)

Table 1. (continued)

RIGHT LOC	V5/MT+, area V5/MT+	48, -70, -1	0.026	0.006	4.139	0.000	0.004	0.024	0.006	3.933	0.000	0.004
RIGHT LOC	OPC, occipital polar cortex	22, -97, 4	0.025	0.010	2.573	0.015	0.067	0.030	0.010	3.128	0.003	0.013
RIGHT LOC	iOccG, inferior occipital gyrus	32, -85, -12	0.028	0.009	3.018	0.005	0.033	0.029	0.009	3.334	0.001	0.010

Figure 8. **Unmodeled within- and between-participant variance in cRSA for mnemonic representations.**

A) Top: Visualization of the main memory task. Participant recognition and judgement confidence was tested for 144 concepts (114 "old", 30 "new"). Bottom: Stacked bar plots with the number of "Hit" (blue) and "Miss" (gray) trials (y-axes) for each participant (x-axis). Participants were sorted by their hit to miss ratio in descending order. **B)** Density plots of cRSA representational strength (y-axes) across the 26 ROIs (x-axes) for Hits (blue) and Misses (gray) for each participant, sorted by descending Hit-to-Miss count ratio. The Hit-to-Miss ratios were displayed in the top-left corner of each subplot, with the font color transitioning from blue (indicating more Hits) to gray (indicating more Misses). **C)** Scatter plot illustrating the influence of the Hit-to-Miss ratio (x-axis) on the participant CVFs (y-axis). The Spearman's correlation was significant, with a coefficient of 0.45 and a p-value of 0.016. Two participants are marked × and were excluded from this correlation as their Hit-to Miss ratio is less than 1. **D)** Density plots (y-axis) of cRSA representational strength across participants for Hits (blue) and Misses (gray). E) Regions exhibiting significant cRSA mnemonic representation (Hits > Misses). Regions highlighted in blue were significant after FDR correction with q < 0.05. Regions in gray were significant at p < 0.05 but did not survive FDR correction.



This issue of neglected heteroscedasticity remains when conditional contrasts (e.g., Hit minus Miss) are computed. Given the nonlinear relationship between trial counts and the reliability of correlations (see Figure 2), a more extreme trial imbalance (e.g., 100 Hits and 10 Misses, versus 70 Hits and 40 Misses) would also lead to a less reliable estimate of the difference. To demonstrate this issue in real fMRI data, we computed the CVF for each participant, measuring how much their Hit-Miss contrast in representational strength estimates deviated from the group mean. Indeed, CVFs were much higher for participants with more extreme Hit to Miss trial counts (rho = 0.45, p = 0.016; see Figure 8C). Critically, the two participants with Hit to Miss ratios < 1 were excluded from this correlation (since in those cases we would expect an increased CVF), but their data points are included in the corresponding figure for transparency.

Additionally, the distributions of representational strengths for Hits and Misses across participants were unequal. Since most participants had more Hit trials than Miss trials (Hit-to-Miss Ratio mean = 3.7, median = 3.1), the distribution of representational strength values was broader for misses compared to hits (Figure 8D). Ultimately, results from cRSA revealed significant mnemonic representation in one right LOC region and one right anterior ITC region, with two additional ITC regions showing trend-level significance (Table 2, Figure 8E); these outcomes were revisited in comparison to the tRSA approach below. Together, these results demonstrate that cRSA failed to model the across-subject variance in real fMRI data, which stemmed from the inevitable variability in trial count ratios affecting the reliability of each participant's cRSA contrast estimates. The violation of statistical assumptions challenged the interpretability of standard t-tests in detecting representational regions.

Subject-level variance accounted for by tRSA

We also sought to assess how the tRSA approach would improve the estimates of representational strength for memory success. The mixed-effects models used to test tRSA mnemonic representations supporting retrieval identified four significant mnemonic representation regions: one in the right LOC and three in the right ITC (Table 2, Figure 9A, and Figure 9B). While the t-statistics for tRSA and cRSA showed a correlation across ROIs (Figure 9C), the tRSA approach, which produced trial-level estimates of representational strength, was able to detect mnemonic representation regions that the cRSA approach did not. This result demonstrated the enhanced sensitivity of tRSA in identifying subtle differences in representational strength. Moreover, a region in the LOC, initially considered a mnemonic representation region by the cRSA method — largely due to data from participants with low miss trial counts (Appendix 3 Table 4) — was correctly excluded from being classified as such by tRSA (Figure 9A, Figure 9B). This result highlighted tRSA's improved specificity in the identification of true mnemonic representation regions by reducing potential biases introduced by variances in trial counts that went neglected by cRSA.

Estimations of continuous modulators

Our final goal was to assess the utility of tRSA in testing hypotheses regarding continuous modulators of representational strength. Here, we focused on the effect of item memorability, a stimulus-level variable (see Figure 10A). The tRSA approach successfully detected continuous, stimulus-level modulation of representational strength during Object Perception, when participants were engaged in basic-level naming of color objects displayed on a white background; notably, no memory task was engaged before or during this phase of the experiment. Four regions within the LOC showed significant modulation by Perceptual Memorability, a measure based on average responses in the Perceptual Retrieval task (see Table 3 and Figure 10B). While not the explicit focus of this analysis, the analyses also identified significant effects of the nuisance variables, Conceptual Memorability, and reaction time in five regions of interest (Appendix 3, Table 6) within the LOC. These regions are often associated with processing the constituent categories or visual features of an object (Tyler et al., 2013), and consistent with the view that the memorability of images can be predicted by statistical properties of semantic features (Bainbridge, 2022; Hovhannisyan et al., 2021; Kramer et al., 2023). Critically, memorability is a robust stimulus property that is consistent across participants and paradigms(Bainbridge, 2022).

			cRSA					tRSA				
REGION	SubArea	Coordinates	Mean	SE	t Values	p	q	coeff	SE	t Values	p	q

Table 2. **Mnemonic representation regions identified by cRSA and tRSA.**

		(x, y, z)										
LEFT IPL	A40c, caudal area 40(PFm)	-56, -49, 38	0.011	-0.012	0.967	0.341	0.617	-0.003	0.005	-0.583	0.560	0.661
RIGHT IPL	A40c, caudal area 40(PFm)	57, -44, 38	0.006	-0.002	0.325	0.747	0.747	0.002	0.005	0.445	0.656	0.742
LEFT IPL	A39rv, rostroventral area 39(PGa)	-47, -65, 26	0.020	-0.011	1.418	0.167	0.546	0.008	0.005	1.842	0.066	0.213
RIGHT IPL	A39rv, rostroventral area 39(PGa)	53, -54, 25	0.007	-0.015	1.048	0.303	0.607	0.005	0.005	1.024	0.306	0.612
LEFT ITC	A20iv, intermediate ventral area 20	-45, -26, -27	0.010	-0.014	1.250	0.221	0.546	0.004	0.005	0.764	0.445	0.661
LEFT ITC	A20r, rostral area 20	-43, -2, -41	-0.022	-0.044	0.894	0.379	0.617	0.005	0.005	1.081	0.280	0.606
LEFT ITC	A20il, intermediate lateral area 20	-56, -16, -28	0.003	-0.011	0.498	0.623	0.704	-0.003	0.005	-0.599	0.549	0.661
RIGHT ITC	A20iv, intermediate ventral area 20	46, -14, -33	0.002	-0.011	0.555	0.584	0.690	**0.013**	**0.005**	**2.678**	**0.007**	**0.048**
RIGHT ITC	A20r, rostral area 20	40, 0, -43	0.015	-0.051	3.289	0.003	0.039	**0.014**	**0.005**	**2.704**	**0.007**	**0.048**
RIGHT ITC	A20il, intermediate lateral area 20	55, -11, -32	0.003	-0.020	1.203	0.239	0.546	0.009	0.005	1.886	0.059	0.213
LEFT ITC	A37elv, extreme lateroventral area37	-51, -57, -15	0.024	0.007	0.903	0.374	0.617	0.001	0.005	0.285	0.776	0.841
LEFT ITC	A37vl, ventrolateral area 37	-55, -60, -6	0.012	-0.007	0.825	0.416	0.636	-0.006	0.005	-1.272	0.203	0.481
LEFT ITC	A20cl, caudolateral of area 20	-59, -42, -16	0.014	0.030	-0.727	0.473	0.683	0.004	0.005	0.783	0.434	0.661
LEFT ITC	A20cv, caudoventral of area 20	-55, -31, -27	0.021	-0.016	1.582	0.125	0.541	0.001	0.005	0.147	0.883	0.918
RIGHT ITC	A37elv, extreme lateroventral area37	53, -52, -18	0.006	-0.019	1.265	0.216	0.546	0.009	0.005	1.978	0.048	0.213
RIGHT ITC	A37vl, ventrolateral area 37	54, -57, -8	0.001	-0.013	0.664	0.512	0.684	0.003	0.005	0.663	0.508	0.661
RIGHT ITC	A20cl, caudolateral of area 20	61, -40, -17	0.008	-0.036	2.137	0.041	0.268	0.008	0.005	1.767	0.077	0.223
RIGHT ITC	A20cv, caudoventral of area 20	54, -31, -26	0.013	-0.031	2.353	0.026	0.222	**0.014**	**0.005**	**3.043**	**0.002**	**0.048**

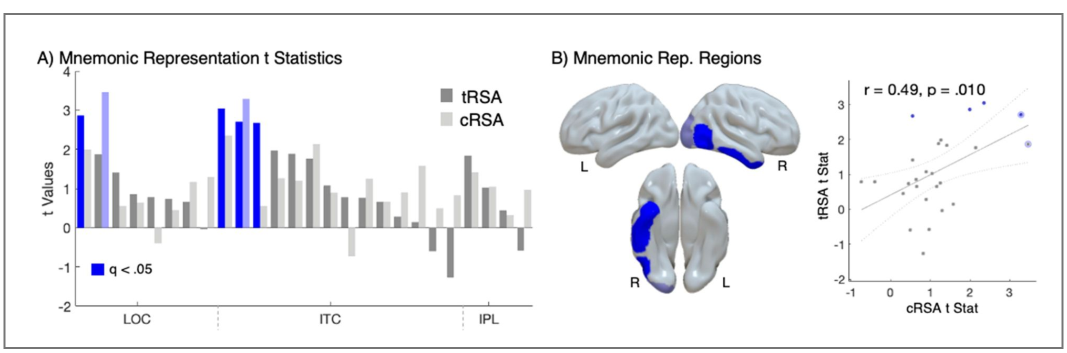
Table 2. (continued)

Table 2. (continued)

LEFT LOC	mOccG, middle occipital gyrus	-31, -89, 11	0.030	-0.001	1.169	0.252	0.546	0.003	0.004	0.661	0.509	0.661
LEFT LOC	V5/MT+, area V5/MT+	-46, -74, 3	0.022	0.013	0.451	0.655	0.710	0.003	0.005	0.740	0.459	0.661
LEFT LOC	OPC, occipital polar cortex	-18, -99, 2	0.028	-0.004	1.294	0.206	0.546	0.000	0.004	-0.035	0.972	0.972
LEFT LOC	iOccG, inferior occipital gyrus	-30, -88, -12	0.001	0.010	-0.393	0.697	0.725	0.004	0.005	0.783	0.433	0.661
RIGHT LOC	mOccG, middle occipital gyrus	34, -86, 11	0.016	0.000	0.642	0.526	0.684	0.004	0.004	0.860	0.390	0.661
RIGHT LOC	V5/MT+, area V5/MT+	48, -70, -1	0.009	-0.037	1.995	0.056	0.289	**0.013**	**0.005**	**2.867**	**0.004**	**0.048**
RIGHT LOC	OPC, occipital polar cortex	22, -97, 4	**0.026**	**-0.062**	**3.461**	**0.002**	**0.039**	0.008	0.005	1.875	0.061	0.213
RIGHT LOC	iOccG, inferior occipital gyrus	32, -85, -12	0.005	-0.004	0.556	0.583	0.690	0.006	0.004	1.411	0.158	0.412

Figure 9. tRSA outperforms cRSA in detecting mnemonic representations.

A) Bar graph of t Statistics for each of the 26 regions of interest. Mnemonic representation t-statistics (y-axis) from cRSA (light) and tRSA (dark) across 26 regions of interest (x-axis). Regions that are significant after FDR correction (q < 0.05) are highlighted in blue. **B)** Brain regions with significant mnemonic representation identified by tRSA are shown in blue. The region that was significant in cRSA but not in tRSA is depicted in light blue. **C)** Scatter plot of t-statistics calculated from tRSA (y-axis) and cRSA (x-axis), showing a significant Pearson correlation (r = 0.49, p = 0.010). Significant tRSA regions are highlighted in blue. Significant cRSA regions are outlined in light blue.



Moreover, object memorability effects have been replicated using a variety of methods aside from tRSA, including univariate analyses and representational analyses of neural activity patterns where trial-level neural activity pattern estimates are correlated directly with object memorability (Slayton et al., 2025).

Summary of Experiment 2

Three key takeaways from Experiment 2 mirrored those from Experiment 1. First, while access to ground truths was not possible with empirical fMRI datasets, results from tRSA and cRSA approaches showed a relatively high degree of correspondence as before (Figure 7C).

Nonetheless, modeling trial-level estimates from tRSA showed improved sensitivity to neural representations in the Object Perception dataset, as the tRSA model was able to weigh study subjects with different numbers of trials appropriately (Figure 7D). Second, in the Memory Retrieval dataset, participants had unbalanced and heterogenous distributions of trial counts in two conditions (“Hits” and “Misses”), as expected with any psychological task focusing on subject performance (Figure 8A). tRSA’s improved characterization can be seen in multiple empirical outcomes: tRSA was able to detect mnemonic representation regions that the cRSA approach missed (improved; sensitivity Figure 9B), and afforded more appropriate weighting of participant’s contrast estimates while maintaining similar outcome statistics as cRSA (Figure 9C). Third, tRSA supported the investigation of research questions that cannot be readily addressed with cRSA, namely the trial- or stimulus-level modulations of representational strength. We demonstrated this point using Item Memorability — a stable stimulus property that is thought to contribute to memory independently of task context, experimental context, or individual differences. Representational strength in several regions in the ventral stream varied continuously with stimulus-level memorability (Figure 10B). Moreover, model-fits became the strongest when the models also incorporated trial-level variables such as fMRI run and reaction time (Appendix 3 Table 6). These findings showed that the representational strength estimates produced by tRSA indeed captured the multi-level variance structure in the data and that tRSA can be implemented to study item-specific or trial-level modulators of representational strength effectively.

Discussion

This paper formally presents tRSA — a novel technique for evaluating the strength of neural representation at the level of individual experimental trials. The performance of tRSA was evaluated and compared to that of cRSA using both simulations (Experiment 1) and empirical fMRI datasets (Experiment 2). Three principal insights can be drawn from these analyses. First, tRSA produces highly similar estimates of *overall* representational strength as cRSA such this new approach comes with little cost in efficacy. Second, tRSA is robust to subject-level variances that would be neglected by cRSA, namely the trial count differences across subjects. With both simulations and empirical datasets, tRSA demonstrated enhanced sensitivity over cRSA in detecting true effects. Third, tRSA provides an opportunity to examine continuous modulators of representational strength with decent specificity and sensitivity. With the empirical fMRI dataset, we demonstrated that tRSA was able to detect significant associations between representational strength and item-level memorability, as well as trial-level nuisance variables. Below we discuss the methodological and conceptual implications of the tRSA approach for the field of cognitive neuroscience and beyond.

Similarities between cRSA and tRSA

Since the advent of cRSA, the method has been widely implemented in studies of neural representations with great success (Dimsdale-Zucker & Ranganath, 2018; Kriegeskorte et al., 2008). Our first and foremost objective was to ensure that our novel tRSA technique produces estimates of representational strength that are comparable to cRSA. This objective was achieved in both Experiment 1 (simulations) and Experiment 2 (real fMRI data). Simulations suggested that tRSA and cRSA estimates reacted in very similar ways to manipulations of the number of subjects, trial counts, and noise level. Although tRSA and cRSA both aim to quantify representational

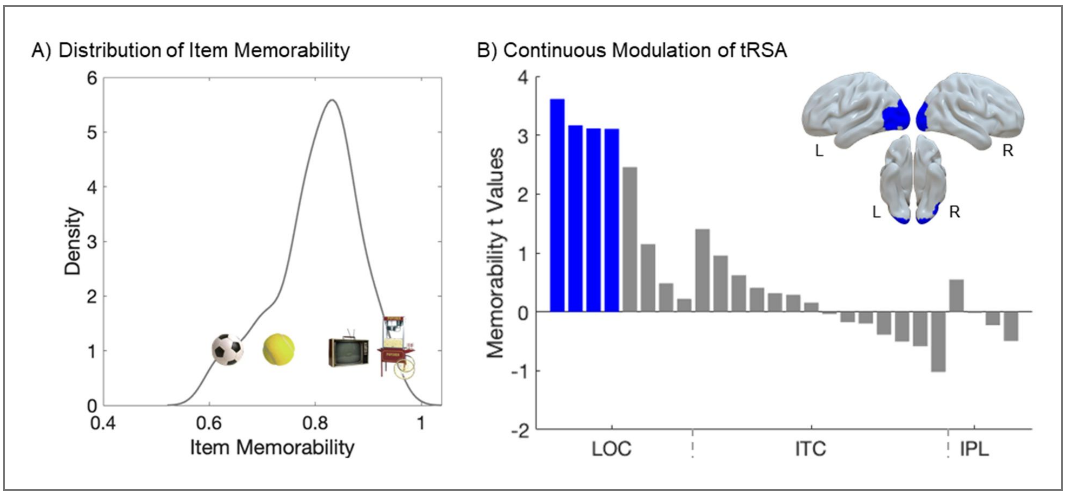


Figure 10. **tRSA captures stimulus-level modulations on representation.**

A) Distribution of Item Memorability. Density plot of memorability values demonstrates a wide distribution of memorability scores (i.e., the average confidence of remembering a particular item). This stimulus-level variance is an underexplored target of representational analyses. **B)** Cortical representational effects of Memorability. Relationship between item-level Memorability and representational strength, as assessed with tRSA, for each of the 26 regions of interest. Trial-level representational strength estimates from Object Perception were significantly modulated by a continuous, stimuluslevel measure of Item Memorability in four regions (blue). The t-statistics for this modulation were plotted (y-axis) across regions of interest (x-axis).

REGION	SubArea	Coordinates (x, y, z)	coeff	SE	t Values	p	q
LEFT IPL	A40c, caudal area 40(PFm)	-56, -49, 38	-0.036	0.072	-0.496	0.620	0.949
RIGHT IPL	A40c, caudal area 40(PFm)	57, -44, 38	0.038	0.069	0.547	0.584	0.949
LEFT IPL	A39rv, rostroventral area 39(PGa)	-47, -65, 26	-0.015	0.068	-0.227	0.820	0.949
RIGHT IPL	A39rv, rostroventral area 39(PGa)	53, -54, 25	-0.001	0.071	-0.018	0.986	0.986
LEFT ITC	A20iv, intermediate ventral area 20	-45, -26, -27	0.044	0.071	0.621	0.534	0.949
LEFT ITC	A20r, rostral area 20	-43, -2, -41	0.025	0.079	0.315	0.752	0.949
LEFT ITC	A20il, intermediate lateral area 20	-56, -16, -28	0.068	0.072	0.955	0.340	0.949
RIGHT ITC	A20iv, intermediate ventral area 20	46, -14, -33	-0.013	0.074	-0.177	0.860	0.949
RIGHT ITC	A20r, rostral area 20	40, 0, -43	-0.046	0.078	-0.587	0.558	0.949

Table 3. Continuous modulation in tRSA.

RIGHT ITC	A20il, intermediate lateral area 20	55, -11, -32	0.021	0.071	0.290	0.771	0.949
LEFT ITC	A37elv, extreme lateroventral area37	-51, -57, -15	0.029	0.071	0.409	0.682	0.949
LEFT ITC	A37vl, ventrolateral area 37	-55, -60, -6	0.105	0.075	1.405	0.160	0.694
LEFT ITC	A20cl, caudolateral of area 20	-59, -42, -16	-0.014	0.068	-0.201	0.841	0.949
LEFT ITC	A20cv, caudoventral of area 20	-55, -31, -27	-0.027	0.069	-0.389	0.698	0.949
RIGHT ITC	A37elv, extreme lateroventral area37	53, -52, -18	-0.073	0.071	-1.022	0.307	0.949
RIGHT ITC	A37vl, ventrolateral area 37	54, -57, -8	0.011	0.070	0.156	0.876	0.949
RIGHT ITC	A20cl, caudolateral of area 20	61, -40, -17	-0.003	0.072	-0.038	0.970	0.986
RIGHT ITC	A20cv, caudoventral of area 20	54, -31, -26	-0.036	0.072	-0.506	0.613	0.949
LEFT LOC	mOccG, middle occipital gyrus	-31, -89, 11	**0.255**	**0.080**	**3.166**	**0.002**	**0.012**
LEFT LOC	V5/MT+, area V5/MT+	-46, -74, 3	**0.258**	**0.071**	**3.614**	**0.000**	**0.008**
LEFT LOC	OPC, occipital polar cortex	-18, -99, 2	**0.276**	**0.089**	**3.113**	**0.002**	**0.012**
LEFT LOC	iOccG, inferior occipital gyrus	-30, -88, -12	0.019	0.083	0.223	0.823	0.949
RIGHT LOC	mOccG, middle occipital gyrus	34, -86, 11	0.201	0.082	2.456	0.014	0.073
RIGHT LOC	V5/MT+, area V5/MT+	48, -70, -1	0.089	0.077	1.149	0.251	0.931
RIGHT LOC	OPC, occipital polar cortex	22, -97, 4	**0.277**	**0.089**	**3.107**	**0.002**	**0.012**
RIGHT LOC	iOccG, inferior occipital gyrus	32, -85, -12	0.040	0.083	0.485	0.628	0.949

Table 3. (continued)

strength, they differ in how they operationalize this concept. cRSA summarizes the correspondence between RSMs as a single measure, such as the matrix-wise Spearman correlation. In contrast, tRSA computes such correspondence for each trial, enabling estimates at the level of individual observations. This flexibility allows trial-level variability to be modeled directly, but also introduces subtle differences in what is being measured. Nonetheless, our simulations showed that, although numerical differences occasionally emerged—particularly when comparing between-condition tRSA estimates to within-condition cRSA estimates—the magnitude of divergence was small and did not affect the outcome of downstream statistical tests. Analyses with real fMRI datasets corroborated their correspondence, showing highly similar distributions of estimated representational strength across the brain. These results confirm that our novel tRSA approach performs at least no worse than cRSA. In other words, researchers can confidently replace their existing cRSA analyses with the tRSA framework, not having to be concerned about the loss of information or inferior statistical performance. It is also important to note that some prior work has examined similarly fine-grained representations in time-resolved neuroimaging data, such as the temporal generalization method introduced by King et al. (King et al., 2019). Their approach trains classifiers at each time point and tests them across all others, resulting in a temporal generalization matrix that reflects decoding accuracy over time. While such matrices share some structural similarity with RSMs, they do not involve correlating trial-level pattern vectors with model RSMs nor do their second-level models include trial-wise, subject-wise, and item-wise variability simultaneously. In general, our analyses with both simulations and real fMRI datasets demonstrated several important advantages of tRSA over cRSA in a wide range of scenarios.

Methodological significance

Statistically, tRSA is designed to properly capture the multi-level variance structure in the data, yielding improved specificity and sensitivity. An important motivation of ours for devising this novel tRSA technique accords with a steady emergence in many scientific fields that advocates for proper treatments of the multi-level variance structure in the data (Dedrick et al., 2009; Winter & Grice, 2021; Yarkoni, 2022). As laid out in the Introduction, we argue that (at least) four independent sources of variance contribute to how strongly a given neural system represents a single event (e.g., looking at an image of “basketball”): condition-level, subjectlevel, stimulus-level, and trial-level.

Subject-level variance was a major focus of analysis in both Experiments. Research subjects are samples (ideally, random and representative) drawn from some population of interest. Barring quantifiable individual differences that may be relevant to the cognitive functions being studied (e.g., age, education, personality traits), subjects are considered to vary randomly around the population-level mean, which is often the focus of analysis. In cRSA, subject-level estimates of representational strength or condition differences are typically entered into a general linear model where homoscedasticity is assumed (e.g., t-test). However, this assumption is easily violated due to heterogeneity in subject-level trial counts, resulting in unreliable model outputs. Indeed, artificially induced variability in trial counts led to catastrophic reductions in the sensitivity of cRSA (Experiment 1). Empirical studies that select trials based on subjective performance or expect to have extensive missingness or exclusions must be wary of this problem, as we demonstrated the same issue of cRSA using real fMRI data from an episodic memory study (Experiment 2). Critically, tRSA proved to be more robust to this subjectlevel heteroscedasticity issue in both Experiments, for trial count information was available to the subsequent models with subjects entered as random factors (Yarkoni, 2022).

Stimulus-level variance was another focus of our investigations. A major advantage of RSA over multivariate pattern classification techniques is that it allows the use of “condition-rich” experimental designs with large and diverse collections of stimuli, so long as meaningful hypotheses can be constructed in the form of an RSM_{model} (Kriegeskorte et al., 2008). For instance, using an RSM_{model} based on human similarity judgments, past work has implemented RSA on neuroimaging data collected with nearly two thousand unique natural object images

(THINGS database; Hebart et al., 2023). Various stimulus properties have also been the target of neural representation research, such as the real-world size of objects (T. Huang et al., 2022), the geographical location of landmarks (Morton et al., 2021), or more abstractly, visual and semantic features of naturalistic images (Devereux et al., 2018; Jozwik et al., 2023; Naspi et al., 2021). Despite such prevalent appreciation for the neurocognitive relevance of stimulus properties, cRSA often does not account for the fact that the same stimulus (e.g., "basketball") is seen by multiple subjects and produces statistically dependent data an issue addressed by Schütt et al. (2023), who developed cross validation and bootstrap methods that explicitly model dependence across both subjects and stimulus conditions. With tRSA, this issue is at least partially addressed by modeling stimulus identity as random effects, resulting in generalizability and reliability (Chen et al., 2021; Yarkoni, 2022). Admittedly, a complete resolution of this stimulus-as-fixed-effect fallacy would require stimulus information to be incorporated in any preceding procedures, including first-level models of fMRI data (Westfall et al., 2017); while beyond the scope of the current paper, these relevant analytical pipelines may be integrated for a full solution of the fallacy. Furthermore, as demonstrated in Experiment 2, tRSA supports the straightforward investigation of how representational strength varies with stimulus-level properties such as memorability. With the flexibility of tRSA, future studies may freely explore other interesting stimulus-level research questions.

Of note, in the current paper, we only examined experimental designs where each unique stimulus is presented once, i.e., no repetitions. Repetitions are sometimes included as targets to which participants should respond, in order to promote engagement level (Allen et al., 2022; Chang et al., 2019; Lahner et al., 2024). Additionally, some preprocessing or modeling procedures use repetitions to improve the accuracy of their estimates of neural activity responses and representations (Charest et al., 2018; Prince et al., 2022), though the effectiveness of this approach may depend on various other factors (Ritchie et al., 2021). Importantly, repetitions have been found to robustly induce neural activity adaptation or *repetition suppression* in task-relevant brain regions (Barron et al., 2016; Grill-Spector et al., 2006), and this phenomenon has profound impacts on multi-voxel activity patterns as well (Mazurchuk et al., 2023). While beyond the scope of the current paper, we briefly mention two possible treatments of stimulus repetitions in implementing tRSA. For one, repetitions can be collapsed in first-level models of brain recordings for more reliable estimations of stimulusspecific neural activations; in this case, any subsequent analyses including tRSA can simply proceed as if individual stimuli are only presented once. Alternatively, repetitions can also be kept as separate events for which individual tRSA estimates are assigned; this option allows one to probe the stability of neural representations across time and repetitions. In either case, our tRSA approach can flexibly suit the various goals of research.

Trial-level variance refers to the effect on the estimations of neural representations by factors that vary from trial to trial. For example, attention level fluctuates during the task and changes the fidelity of neural representations (Aly & Turk-Browne, 2016; Rothlein et al., 2018). Like stimulus-level variance, trial-level variance is not accounted for by cRSA as the comparison of RSMs collapses across trials. In our simulations, we demonstrated that tRSA can indeed capture trial-level variance and reliably estimate its effect on representational strength (Experiment 1). In a real fMRI dataset, we found that reaction time, a trial-level nuisance variable, also had statistically significant effects on representational strength (Experiment 2). In addition, past research has found that pure physiological changes in pupillary size, cardiac rhythm, and respiration may also influence both neural and cognitive processes (Critchley & Garfinkel, 2018; van der Wel & van Steenbergen, 2018). While a portion of those effects could be due to the incomplete removal of artifacts during signal preprocessing, some physiological changes may also have intriguing neurocognitive relevance that is worth exploring in future studies.

Conceptual significance

Fundamentally, approaching RSA at the level of individual trials is an advancement in terms of not only *methodology* (discussed above) but also *conceptualization*. The notion of *neural representation*, albeit used more broadly than the phrase is intended to, emphasizes the link between the representee (contents being represented, such as stimuli or memories) and its representative (alternative format of the content, such as neural activity) (Baker et al., 2022; Favela & Machery, 2023; Vilarroya, 2017). Collapsing trial-level information in its formulation, cRSA could only indicate the quality of a *kind* of representation or the *overall* representational strength across events. Instead, tRSA provides a measure that directly corresponds to the conceptualization of representation, with each trial or event receiving its own estimated representational strength. This level of analysis is consistent with previous studies that focused on quantifying the similarity of multi-voxel activity patterns across different trials (S. Huang, Faul, et al., 2024; Wing et al., 2020; Yu et al., 2024) and at different stages of memory (Ritchey et al., 2013; Shao et al., 2023). As such, tRSA provides a versatile tool for inquiring about many different brain-behavior relationships at an appropriate level of analysis and with straightforward interpretations (Becker et al., 2024; Howard et al., 2024; Morales-Torres et al., 2024; Naspi et al., 2023; Pacheco-Estefan et al., 2024). Notably, tRSA is not limited to neuroimaging data; past research has also implemented this approach to perform item-specific analyses on behavioral data (Walsh & Rissman, 2023).

At a broader level, the tRSA approach also engenders a host of other innovative techniques. While a number of studies have addressed the validity of measuring representational geometry using designs with multiple repetitions, a conceptual benefit of the tRSA approach is the reliance on a regression framework that engenders the testing of competing *conceptual* models of stimulus representation (e.g., taxonomic vs. encyclopedic semantic features, as in Davis et al., 2021). The emerging research on representational connectivity focuses on the intersection of representational information and functional connectivity analyses, and seeks to uncover new multivariate views on how concepts are distributed across the brain (Anzellotti & Coutanche, 2018). Using the tRSA approach, trial-level representational strength values from any pair of brain regions may be correlated to yield their model-based representational connectivity (S. Huang, De Brigard, et al., 2024; Karimi-Rouzbahani et al., 2022). Trial-level representational strength values may also be combined with other trial-level measures of neural activity such as univariate activation to probe diverse inter-regional interactions that are theoretically motivated (S. Huang, Howard, et al., 2024).

Methodological considerations

While linear mixed-effects modeling offers a powerful framework for analyzing representational similarity measures, it is critical that researchers carefully construct and validate their models. In our simulations, we designed our model RSM to be the "true" RSM for demonstration purposes. However, researchers should always consider if their models match the goals of their analysis, including 1) constructing the random effects structure that will converge in their dataset and 2) testing their model fits against alternative structures (Meteyard & Davies, 2020; Park et al., 2020) and 3) considering which effects should be considered random or fixed depending on their research question. An effect is fixed when the levels represent the specific conditions of theoretical interest (e.g., *task condition*) and the goal is to estimate and interpret those differences directly. In contrast, an effect is random when the levels are sampled from a broader population (e.g., *subjects*) and the goal is to account for their variability while generalizing beyond the sample tested. Note that the same variable (e.g., *stimuli*) may be considered fixed or random depending on the research questions.

Additionally, the multi-level structure of RSA measures introduces potential dependencies across subjects, stimuli, and trials, which can violate assumptions of independence if not properly modeled. In the present study, we used a model that included random intercepts for both subjects and stimuli, which accounts for variance at these levels and improves the generalizability of fixed-effect estimates. Still, there is a potential for systematic dependence across trials within a subject.

To ensure that the model assumptions were satisfied, we conducted a series of diagnostic checks on an exemplar ROI (right LOC; middle occipital gyrus) in the Object Perception dataset, including visual inspection of residual distributions and autocorrelation (Appendix 3, Figure 13). These diagnostics supported the assumptions of normality, homoscedasticity, and conditional independence of residuals. In addition, we conducted permutation-based inference, similar to prior improvements to cRSA (Nili et al., 2014), using a nested model comparison to test whether the mean similarity in this ROI was significantly greater than zero. The observed likelihood ratio test statistic fell in the extreme tail of the null distribution (Appendix 3, Figure 14), providing strong nonparametric evidence for the reliability of the observed effect. We emphasize that this type of model checking and permutation testing is not merely confirmatory but can help validate key assumptions in RSA modeling, especially when applying mixed-effects models to neural similarity data. Researchers are encouraged to adopt similar procedures to ensure the robustness and interpretability of their findings. Further, while analyses here were largely employed to be comparable with cRSA, researchers should consider taking advantage of the flexibility of the mixed-effects models and include co variates of non-interest (run, trial order, etc.).

Additionally, we note that while our proposed tRSA framework provides a flexible and statistically principled approach for modeling trial-level representational strength, we acknowledge that there are alternative methods for addressing trial-level variability in RSA. In particular, the use of cross-validated distance metrics (e.g., crossnobis distance) has become increasingly popular for controlling differences in measurement noise variance and accounting for possible covariance structures across trials (Walther et al., 2016). These metrics offer several advantages, including unbiased estimation of representational dissimilarities under Gaussian noise assumptions and improved generalization to unseen data. However, crossvalidated distances are conceptually distinct from the approach taken here: whereas crossvalidation aims to correct for noise-related biases in representational dissimilarity matrices, our trial-level RSA method focuses on estimating and modeling the variability in representation strength across individual trials using mixed-effects modeling. Rather than proposing a replacement for cross-validated RSA, tRSA adds a complementary tool to the methodological toolkit—one that supports hypothesis-driven inference about condition effects and trial-level covariates, while leveraging the full structure of the data.

Conclusion

We present a comprehensive overview and diagnostics of a trial-level approach to representational similarity analysis, termed tRSA. With both simulations and real fMRI datasets, we have demonstrated that tRSA properly captures the multi-level variance structure in the data, shows improved and robust specificity and sensitivity, and offers flexible modeling options. We believe that tRSA is an important advancement of the generic RSA implementation, both methodologically and conceptually, and we hope that this innovation provides a versatile and useful tool for digging deeper into the fascinating complexity of cognition.

Appendices

Appendix 1. Row-wise correlations

We explored the statistical properties of computing row-wise correlations, which is a critical step in tRSA. Assuming an experimental design with n = 200 trials, we sampled $n(n - 1)/2$ = 19900 pairs of values from a bivariate normal distribution, with their Pearson correlation fixed at some given value via the ‘MASS 7.3-60.2’ package (Venables et al., 2002) in R. The two vectors were rearranged as the lower triangular part of two $n \times n$ matrices, and the upper triangular parts were filled out accordingly to yield two symmetric matrices M and S. With each pair of matrices, the Fisher-transformed Pearson correlation coefficient of the lower triangular parts of the two matrices is equal to the ground truth parameter z. We computed rowwise Pearson correlations for row vectors $M_{i.}$ and $S_{i.}$, barring entries on the diagonal (see Figure 1C, Bottom). Fisher transformation of correlation coefficients was applied such that the distribution of values is more approximately normal (Silver & Dunlap, 1987). This set of procedures was repeated for 10,000 iterations, for each ground truth parameter z ranging from *-0.2* to *0.6* with increments of *0.1*.Our simulations indicated that the average of row-wise correlations numerically approximates the overall correlation. Specifically, the estimates centered around zero when it was the ground truth; as the magnitude of the ground truth increased, there appeared to be a bias toward slightly larger magnitudes. Despite this, the spread of estimates derived from the row-wise method remained small, with a constrained range of [0.599, 0.604] around the ground truth of 0.600 (see Appendix 11 Figure 11).

Note that a limitation of this simulation is that the symmetric matrices were randomly generated without being guaranteed to be mathematically valid similarity matrices (i.e., positive semidefinite). For example, [[1, 1, 1], [1, 1, -1], [1, -1, 1]] is symmetric but is not a valid correlation RSM, since it is impossible to satisfy $a = b = c$ (row 1) and $a = b = -c$ (row 2) unless $a = b = c = 0$, in which case the Pearson correlation is undefined. Similarly, RSMs using distance metrics such as Euclidean distance must satisfy the triangle inequality, i.e., $d_{xy} + d_{xz} > d_{yz}$. Moreover, it is difficult to implement the multi-level variance structure directly on RSMs; rather, those variances are theorized to operate directly on the underlying *activity patterns*.

Appendix 2. Comparing across-condition and within-condition tRSA

Given discrete experimental conditions, tRSA could be formulated either within each condition or across all conditions. In *within-condition* tRSA, one would generate condition-specific RSMs, e.g., one set of $n_A \times n_A$ RSMs for Condition A and another set of $n_B \times n_B$ RSMs for Condition B, and compute the two sets of *within-condition* tRSA values separately. These tRSA estimates are then subjected to subsequent analysis. In *across-condition* tRSA, RSMs are generated across all trials in all conditions, and tRSA estimates are computed as if there were no multiple conditions. Then, tRSA estimates are categorized based on the condition they belong to in any subsequent analysis.

We advocate for the use of across-condition tRSA for its enhanced statistical reliability over within-condition tRSA. One major limitation of within-condition tRSA is its vulnerability to small and variable trial counts. Suppose that a condition contains n trials, cRSA would use $n(n - 1)/2$ observations to compute its estimate, whereas within-condition tRSA could only use $n - 1$ observations. This effective sample size could turn out to be fairly small and variable in practice (see Figure 8A, Bottom), resulting in unreliable estimates (see Figure 2). On the contrary, across-condition tRSA suffers much less from this issue, since most experiments should have been designed to have an adequate total trial count.

Another theoretical advantage of across-condition tRSA is that it better corresponds to $\text{tRSA}_{\text{continuous}}$, i.e., analyses in which trial-level representational strength estimates are hypothesized to be modulated by some continuous variable such as memorability or reaction time. In both across-condition tRSA and $\text{tRSA}_{\text{continuous}}$, full RSMs are used to estimate representational strength before any discrete condition effect or continuous modulation effect is assessed.

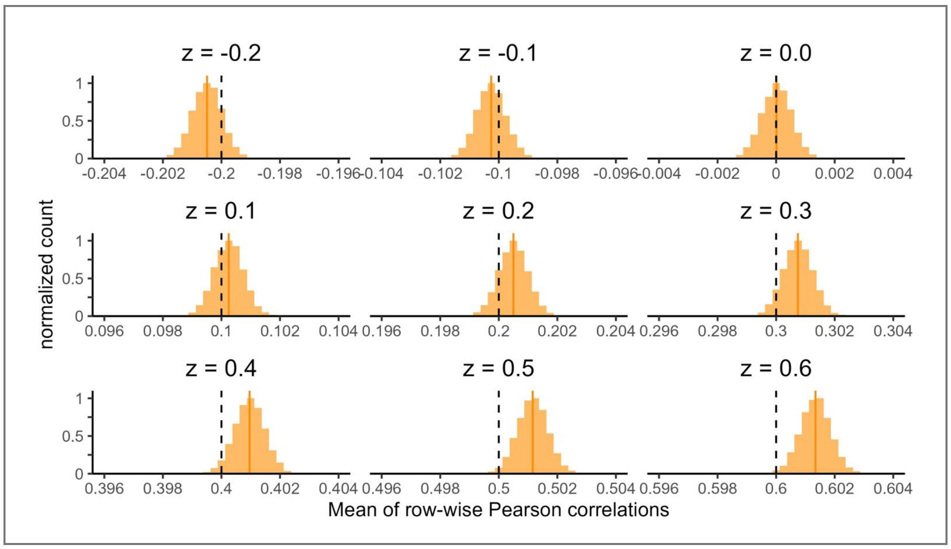
Appendix 2 Figure 12 below summarizes the differences between within-condition and across-condition tRSA outcomes in a series of simulations. Notably, while within-condition tRSA estimates showed better numerical correspondence with cRSA estimates, they are less reliable than across-condition tRSA, as evidenced by the wider spread of estimates.

Appendix 3. Supporting data and results for Experiment 2

Exemplar Permutation Testing

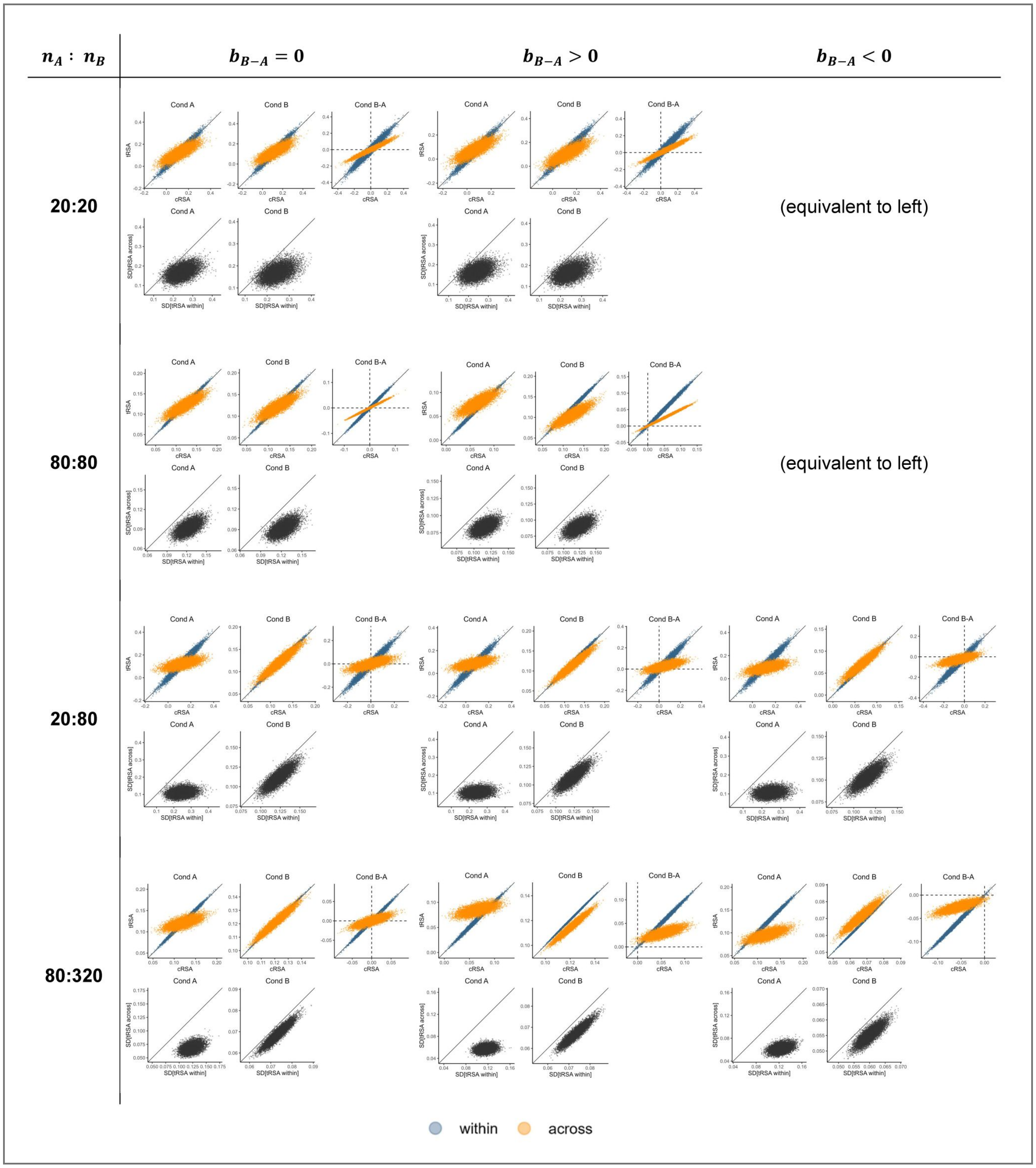
To test whether the mean representational strength in the ROI right LOC (middle occipital gyrus) was significantly greater than zero, we used a permutation-based likelihood ratio test implemented via the ‘permlmer’ function. This test compares two nested linear mixedeffects models fit using the ‘Imer’ function from the lme4 package, both including random intercepts for Participant and Stimulus ID to account for between-subject and between-item variability.

The null model excluded a fixed intercept term, effectively constraining the mean similarity to zero after accounting for random effects:



Appendix 1 Figure 11. **Row-wise correlations.**

Histograms of the mean of row-wise correlation values. Each histogram depicted 10,000 iterations. In each iteration, two symmetrical matrices with a fixed ground truth Pearson correlation coefficient were randomly generated, and the mean of row-wise correlation coefficients were computed as the estimate. Dashed vertical lines indicate the ground truth parameter, and solid vertical lines indicate the mean of estimates.



Appendix 2 Figure 12. Comparing across-condition and within-condition tRSA.

Each cell depicts descriptive statistics of different RSA methods from 10,000 simulations. Cells differ in raw trial counts (20, 80, or 320 in a given condition), balance of trial counts between conditions (balanced or unbalanced), and effect (A = B, A > B, or A < B). In each cell, the top scatter plots depict condition-level tRSA estimates (y-axis; blue, within-condition; orange, across-condition) against cRSA estimates (x-axis), whereas the bottom scatter plots depict the standard deviations of trial-level representational strength estimates from across-condition tRSA (y-axis) and within-condition tRSA (x-axis).

$$ROI \sim 0 + (1|Participant) + (1|Stimulus)$$

The full model included the same random effects structure but allowed the intercept to be freely estimated:

$$ROI \sim 1 + (1|Participant) + (1|Stimulus)$$

By comparing the fit of these two models, we directly tested whether the average similarity in this ROI was significantly different from zero. Permutation testing (1,000 permutations) was used to generate a nonparametric p-value, providing inference without relying on normality assumptions.

The full model, which estimated a nonzero mean similarity in the right LOC (middle occipital gyrus), showed a significantly better fit to the data than the null model that fixed the mean at zero ($x^2(1) = 17.60$, $p = 2.72 \times 10^{-5}$). The permutation-based p-value obtained from permlmer confirmed this effect as statistically significant ($p = 0.0099$), indicating that the mean similarity in this ROI was reliably greater than zero. These results support the conclusion that the right LOC contains representational structure consistent with the HMAXc2 RSM. A density plot of the permuted likelihood ratio tests is plotted along with the observed likelihood ratio test in Appendix 3 Figure 14.

Data availability

Data is available upon request to the corresponding author and our simulations and example tRSA code is available at *https://github.com/electricdinolab*.

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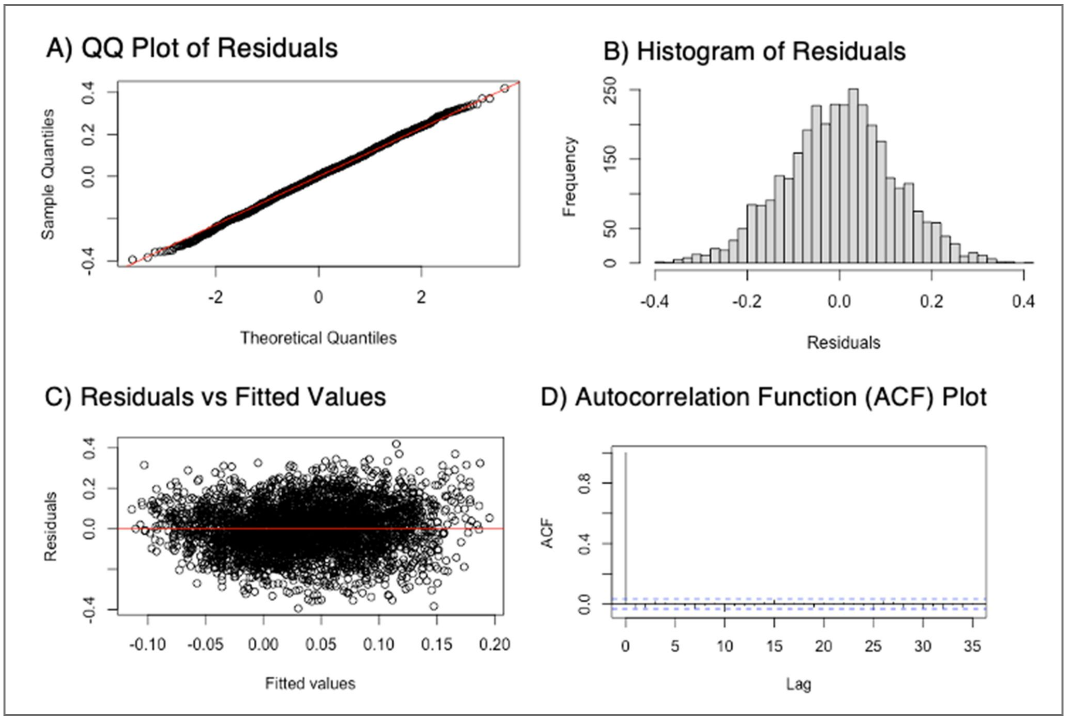
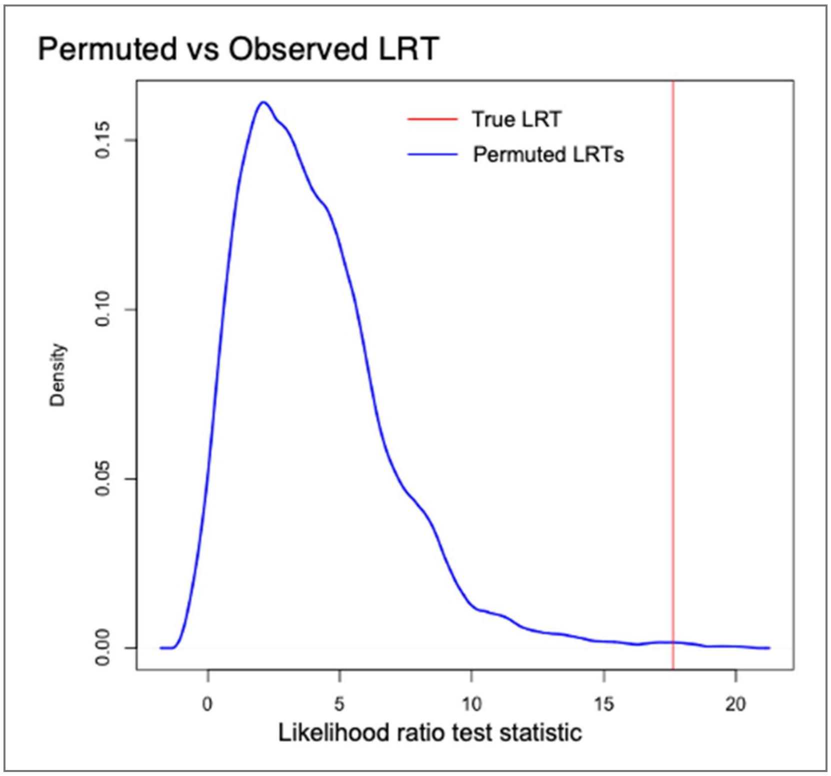


Appendix 3 Figure 13. Exemplar model checks Experiment 2 Object Perception in right LOC (middle occipital gyrus).

A) QQ Plot of Residuals. **B)** Histogram of Residuals. **C)** Residuals vs Fitted Values. **D)** Autocorrelation Function (ACF) Plot. These diagnostic plots were generated to evaluate whether the linear mixed-effects model satisfied key statistical assumptions. The model was fit to an exemplar region of interest (ROI) with subject and stimulus modeled as random intercepts. Residuals were extracted from the fitted model and plotted using base R functions. The QQ plot (A) and histogram (B) indicate that residuals are approximately normally distributed. The residuals vs. fitted values plot (C) shows no clear patterns or funneling, suggesting homoscedasticity and no evidence of model misspecification. Finally, the ACF plot (D) displays no significant autocorrelation, supporting the assumption that residuals are independent after accounting for the specified random effects structure.



Appendix 3 Figure 14. Permutation-based likelihood ratio test (LRT) comparing models with and without a fixed intercept in the right LOC (middle occipital gyrus).

The x-axis shows the distribution of LRT statistics obtained from 1,000 permutations under the null hypothesis that the mean similarity is zero. The blue line represents the null distribution of permuted LRT values, and the red vertical line indicates the observed LRT statistic from the original model comparison ($X^2 = 17.60$). The observed value falls in the extreme tail of the null distribution, yielding a permutation-based p-value of 0.0099, indicating that the mean similarity in this ROI is significantly greater than zero."

Participant	Gender	Response Trials at Perception	ROC	Count of Hits	Hit Rate
1	M	114	0.84	84	0.74
2	W	114	0.73	82	0.78
3	**W**	105	**0.70**	**32**	**0.36**

Appendix 3 Table 4. **Behavioral results**

Bold indicates participants excluded from Conceptual Retrieval Analyses.

4	W	112	0.93	94	0.82
5	W	114	0.95	85	0.76
6	W	96	0.74	84	0.79
7	W	108	0.93	99	0.88
8	W	114	0.56	76	0.75
9	M	113	0.81	89	0.78
10	M	101	0.79	64	0.80
11	M	101	0.93	78	0.80
12	W	113	0.66	53	0.50
13	M	114	0.87	89	0.78
14	W	103	0.69	35	0.39
15	M	114	0.84	75	0.66
16	W	114	0.67	51	0.45
17	**W**	**113**	**0.35**	**26**	**0.30**
18	W	114	0.83	75	0.66
19	W	114	0.90	100	0.88
20	W	114	0.71	68	0.60
21	W	114	0.89	80	0.70
22	W	114	0.90	99	0.89
23	W	113	0.83	93	0.82
24	W	110	0.70	64	0.62
25	M	114	0.87	84	0.74
26	M	95	0.62	71	0.75
27	M	114	0.75	81	0.71
28	M	109	0.95	100	0.88
29	W	106	0.72	81	0.79
30	W	114	0.92	95	0.86

Appendix 3 Table 4. (continued)

31	M	114	0.74	79	0.69
32	W	109	0.62	50	0.50

Appendix 3 Table 4. (continued)

Appendix 3 Table 5. **Model selection using 3 criteria: Akaike information criterion (AIC), Bayesian information criterion (BIC), and log-liklihood ratio testing (LRT).**

			(1+adjHvM|Subject + (1+adjHvM|StimID)			(1+adjHvM|Subject)+(1|StimID)			(1|Subject)+(1|StimID)		
REGION	SubArea	Coordinates (x, y, z)	AIC	BIC	LRT pvalue	AIC	BIC	LRT pvalue	AIC	BIC	LRT pvalue
LEFT IPL	A40c, caudal area 40(PFm)	-56, -49, 38	-4778	-4724	-	-4781	-4739	0.555	-4784	-4754	0.600
RIGHT IPL	A40c, caudal area 40(PFm)	57, -44, 38	-4632	-4578	-	-4614	-4571	0.000	-4639	-4608	1.000
LEFT IPL	A39rv, rostroventral area 39(PGa)	-47, -65, 26	-4907	-4852	-	-4910	-4867	0.674	-4914	-4883	0.819
RIGHT IPL	A39rv, rostroventral area 39(PGa)	53, -54, 25	-4765	-4710	-	-4768	-4726	0.909	-4772	-4742	0.969
LEFT ITC	A20iv, intermediate ventral area 20	-45, -26, -27	-4786	-4731	-	-4790	-4747	1.000	-4788	-4758	0.079
LEFT ITC	A20r, rostral area 20	-43, -2, -41	-4188	-4134	-	-4187	-4145	0.117	-4191	-4161	0.858
LEFT ITC	A20il, intermediate lateral area 20	-56, -16, -28	-4808	-4753	-	-4806	-4764	0.064	-4809	-4779	0.663
RIGHT ITC	A20iv, intermediate ventral area 20	46, -14, -33	-4526	-4472	-	-4529	-4487	0.623	-4533	-4502	0.716
RIGHT ITC	A20r, rostral area 20	40, 0, -43	-3940	-3886	-	-3942	-3901	0.510	-3944	-3914	0.337
RIGHT ITC	A20il, intermediate lateral area 20	55, -11, -32	-4554	-4500	-	-4558	-4516	0.982	-4562	-4532	0.974
LEFT ITC	A37elv, extreme lateroventral area37	-51, -57, -15	-4920	-4865	-	-4924	-4881	1.000	-4924	-4894	0.214
LEFT ITC	A37vl, ventrolateral area 37	-55, -60, -6	-4946	-4891	-	-4949	-4907	0.875	-4953	-4923	0.875
LEFT ITC	A20cl, caudolateral of area 20	-59, -42, -16	-5003	-4949	-	-5005	-4962	0.297	-5007	-4977	0.476
LEFT ITC	A20cv, caudoventral of area 20	-55, -31, -27	-4867	-4812	-	-4871	-4828	0.996	-4873	-4843	0.413
RIGHT ITC	A37elv, extreme lateroventral area37	53, -52, -18	-4933	-4878	-	-4935	-4892	0.301	-4936	-4906	0.352
RIGHT ITC	A37vl, ventrolateral area 37	54, -57, -8	-4875	-4821	-	-4877	-4835	0.366	-4881	-4850	0.865
RIGHT ITC	A20cl, caudolateral of area 20	61, -40, -17	-4617	-4563	-	-4619	-4576	0.291	-4623	-4592	0.966
RIGHT ITC	A20cv, caudoventral of area 20	54, -31, -26	-4683	-4628	-	-4685	-4642	0.293	-4688	-4657	0.679
LEFT LOC	mOccG, middle occipital gyrus	-31, -89, 11	-5123	-5068	-	-5127	-5084	0.941	-5128	-5097	0.259
LEFT LOC	V5/MT+, area V5/MT+	-46, -74, 3	-4820	-4765	-	-4824	-4782	0.996	-4827	-4797	0.631
LEFT LOC	OPC, occipital polar cortex	-18, -99, 2	-5046	-4991	-	-5049	-5006	0.595	-5052	-5022	0.818
LEFT LOC	iOccG, inferior occipital gyrus	-30, -88, -12	-4959	-4904	-	-4962	-4919	0.516	-4966	-4935	0.929
RIGHT LOC	mOccG, middle occipital gyrus	34, -86, 11	-5265	-5210	-	-5269	-5227	1.000	-5271	-5241	0.440
RIGHT LOC	V5/MT+, area V5/MT+	48, -70, -1	-4879	-4824	-	-4881	-4839	0.382	-4883	-4852	0.279
RIGHT LOC	OPC, occipital polar cortex	22, -97, 4	-4995	-4940	-	-4989	-4946	0.007	-4991	-4961	0.472
RIGHT LOC	iOccG, inferior occipital gyrus	32, -85, -12	-5126	-5071	-	-5126	-5083	0.144	-5129	-5098	0.618
		% Selected	4%	0%	8%	0%	0%	0%	96%	100%	92%

			Conceptual Memorability					Reaction Time				
REGION	SubArea	Coordinates (x, y, z)	Mean	SE	t Values	p	q	coeff	SE	t Values	p	q
LEFT IPL	A40c, caudal area 40(PFm)	-56, -49, 38	0.077	0.060	1.274	0.203	0.659	0.005	0.004	1.369	0.171	0.869
RIGHT IPL	A40c, caudal area 40(PFm)	57, -44, 38	-0.053	0.057	-0.931	0.352	0.773	0.007	0.004	1.731	0.084	0.869
LEFT IPL	A39rv, rostroventral area 39(PGa)	-47, -65, 26	-0.019	0.056	-0.341	0.733	0.866	0.005	0.004	1.433	0.152	0.869
RIGHT IPL	A39rv, rostroventral area 39(PGa)	53, -54, 25	-0.039	0.059	-0.666	0.506	0.773	0.010	0.004	2.482	0.013	0.869
LEFT ITC	A20iv, intermediate ventral area 20	-45, -26, -27	-0.052	0.059	-0.876	0.381	0.773	0.004	0.004	1.101	0.271	0.869
LEFT ITC	A20r, rostral area 20	-43, -2, -41	-0.051	0.065	-0.780	0.436	0.773	-0.004	0.004	-0.846	0.397	0.869
LEFT ITC	A20il, intermediate lateral area 20	-56, -16, -28	-0.053	0.060	-0.893	0.372	0.773	0.004	0.004	0.889	0.374	0.869
RIGHT ITC	A20iv, intermediate ventral area 20	46, -14, -33	0.017	0.061	0.278	0.781	0.867	0.002	0.004	0.556	0.578	0.869
RIGHT ITC	A20r, rostral area 20	40, 0, -43	0.029	0.065	0.444	0.657	0.854	0.000	0.004	0.016	0.987	0.869
RIGHT ITC	A20il, intermediate lateral area 20	55, -11, -32	-0.053	0.059	-0.887	0.375	0.773	0.002	0.004	0.439	0.661	0.869
LEFT ITC	A37elv, extreme lateroventral area37	-51, -57, -15	-0.044	0.059	-0.750	0.453	0.773	0.004	0.004	1.050	0.294	0.869
LEFT ITC	A37vl, ventrolateral area 37	-55, -60, -6	-0.043	0.062	-0.682	0.495	0.773	-0.003	0.004	-0.627	0.531	0.869
LEFT ITC	A20cl, caudolateral of area 20	-59, -42, -16	0.009	0.057	0.165	0.869	0.869	0.002	0.004	0.529	0.597	0.869
LEFT	A20cv, caudoventral of	-55, -31, -27	0.032	0.057	0.550	0.582	0.797	0.001	0.004	0.218	0.827	0.869

Appendix 3 Table 6. **Significant nuisance effects in memorability model.**

ITC	area 20											
RIGHT ITC	A37elv, extreme lateroventral area37	53, -52, -18	0.034	0.059	0.580	0.562	0.797	0.000	0.004	0.015	0.988	0.869
RIGHT ITC	A37vl, ventrolateral area 37	54, -57, -8	-0.012	0.059	-0.210	0.834	0.867	-0.007	0.004	-1.662	0.097	0.869
RIGHT ITC	A20cl, caudolateral of area 20	61, -40, -17	0.022	0.060	0.366	0.714	0.866	-0.001	0.004	-0.137	0.891	0.869
RIGHT ITC	A20cv, caudoventral of area 20	54, -31, -26	0.013	0.060	0.221	0.825	0.867	-0.004	0.004	-0.909	0.363	0.869
LEFT LOC	mOccG, middle occipital gyrus	-31, -89, 11	**-0.233**	**0.067**	**-3.473**	**0.001**	**0.005**	**0.015**	**0.005**	**3.199**	**0.001**	**0.033**
LEFT LOC	V5/MT+, area V5/MT+	-46, -74, 3	**-0.193**	**0.059**	**-3.241**	**0.001**	**0.006**	**0.008**	**0.004**	**2.093**	**0.036**	**0.033**
LEFT LOC	OPC, occipital polar cortex	-18, -99, 2	**-0.259**	**0.074**	**-3.516**	**0.000**	**0.005**	**0.017**	**0.005**	**3.359**	**0.001**	**0.033**
LEFT LOC	iOccG, inferior occipital gyrus	-30, -88, -12	-0.054	0.069	-0.779	0.436	0.773	0.011	0.005	2.371	0.018	0.869
RIGHT LOC	mOccG, middle occipital gyrus	34, -86, 11	**-0.224**	**0.068**	**-3.286**	**0.001**	**0.006**	**0.015**	**0.005**	**3.361**	**0.001**	**0.033**
RIGHT LOC	V5/MT+, area V5/MT+	48, -70, -1	-0.089	0.064	-1.388	0.165	0.614	0.006	0.004	1.311	0.190	0.869
RIGHT LOC	OPC, occipital polar cortex	22, -97, 4	**-0.296**	**0.074**	**-3.987**	**0.000**	**0.002**	**0.014**	**0.005**	**2.867**	**0.004**	**0.033**
RIGHT LOC	iOccG, inferior occipital gyrus	32, -85, -12	-0.099	0.069	-1.439	0.150	0.614	0.012	0.005	2.610	0.009	0.869

Appendix 3 Table 6. (continued)

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Peer reviews

Reviewer #2 (Public review):

This paper proposes two changes to classic RSA, a popular method to probe neural representation in neuroimaging experiments: computing RSA at row/column level of RDM, and using linear mixed modeling to compute second level statistics, using the individual row/columns to estimate a random effect of stimulus. The benefit of the new method is demonstrated using simulations and a re-analysis of a prior fMRI dataset on object perception and memory encoding.

The author's claim that tRSA is a promising approach to perform more complete modeling of cogneuro data, and to conceptualize representation at the single trial/event level (cf Discussion section on P42), is appealing.

In their revised manuscript, the authors have addressed some previous concerns, now referencing more literature aiming to improve RSA and its associated statistical inferences, and providing more guidance on methodological considerations in the Discussion. However, I wish the authors had more extensively edited the Introduction to better contextualize the work and clarify the specific settings in which they see the method as being beneficial over classic RSA. For example, some of the limitations of cRSA mentioned on page 6, e.g. related to presenting the same stimuli to multiple subjects, seem to be quite specific to settings where the researcher expects differential responses across subjects to fundamentally alter the interpretation, rather than something that will just average out by repeatedly offering the same stimulus, or combining data across subjects. It's not clear to me how the switch from 'matrix-level' to 'row-level' analysis in tRSA necessarily addresses this problem. I would be very helpful if the authors would more explicitly outline what problem the row-level aspect of tRSA is solving; what problem statistical inference via LMM is solving; and walk the reader through a very specific use case (perhaps a toy version of the real-data experiment which is now at the end of the paper). Explaining the utility of tRSA for experimental settings in which assessing representational strength for a single-events is crucial would clarify the contribution of this new method better.

A few weaknesses mentioned in my previous review were not adequately addressed. To demonstrate the utility of the method on real neural recordings, only a single dataset is used with a quite complicated experimental design; it's not clear if there is any benefit of using tRSA on a simpler real dataset. Moreover, the cells of an RDM/RSM reflect pairwise comparisons between response patterns. Because the response patterns are repeatedly compared, the cells of this matrix are not independent of one another. While the authors show examples that failure to meet independence assumptions do not affect results in their specific dataset, it does not get acknowledged as a problem at a more fundamental level.

Finally, while the paper now states that 'simulations and example tRSA code' are publicly available, the link points to the lab's general github page containing many lab repositories, in which I could not identify a specific repository related to this paper. This is disappointing given that the main goal of this manuscript is to provide a new method that they encourage others to use; a clear pointer to available code is only a minimal requirement to achieve that goal. A dedicated repository, including documentation, READMEs and tutorials/demo's to run simulations, compare methods, etc. would greatly enhance the paper's contribution.

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Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *(1) I have to admit that it took a few hours of intense work to understand this paper and to even figure out where the authors were coming from. The problem setting, nomenclature, and simulation methods presented in this paper do not conform to the notation common in the field, are often contradictory, and are usually hard to understand. Most importantly, the problem that the paper is trying to solve seems to me to be quite specific to the particular memory study in question, and is very different from the normal setting of model-comparative RSA that I (and I think other readers) may be more familiar with.*

We have revised the paper for clarity at all levels: motivation, application, and parameterization. We clarify that there is a large unmet need for using RSA in a trial-wise manner, and that this approach indeed offers benefits to any team interested in decoding trial-wise representational information linked to a behavioral responses, and as such is not a problem specific to a single memory study.

> *(2) The definition of "classical RSA" that the authors are using is very narrow. The group around Niko Kriegeskorte has developed RSA over the last 10 years, addressing many of the perceived limitations of the technique. For example, cross-validated distance measures (Walther et al. 2016; Nili et al. 2014; Diedrichsen et al. 2021) effectively deal with an uneven number of trials per condition and unequal amounts of measurement noise across trials. Different RDM comparators (Diedrichsen et al. 2021) and statistical methods for generalization across stimuli (Schütt et al. 2023) have been developed, addressing shortcomings in sensitivity. Finally, both a Bayesian variant of RSA (Pattern component modelling, (Diedrichsen, Yokoi, and Arbuckle 2018) and an encoding model (Naselaris et al. 2011) can effectively deal with continuous variables or features across time points or trials in a framework that is very related to RSA (Diedrichsen and Kriegeskorte 2017). The author may not consider these newer developments to be classical, but they are in common use and certainly provide the solution to the problems raised in this paper in the setting of model-comparative RSA in which there is more than one repetition per stimulus.*

We appreciate the summary of relevant literature and have included a revised Introduction to address this bounty of relevant work. While much is owed to these authors, new developments from a diverse array of researchers outside of a single group can aid in new research questions, and should always have a place in our research landscape. We owe much to the work of Kriegeskorte's group, and in fact, Schutt et al., 2023 served as a very relevant touchpoint in the Discussion and helped to highlight specific needs not addressed by the assessment of the "representational geometry" of an entire presented stimulus set. Principal amongst these needs is the application of trial-wise representational information that can be related to trial-wise behavioral responses and thus used to address specific questions on brain-behavior relationships. We invite the Reviewer to consider the utility of this shift with the following revisions to the Introduction.

Page 3. “Recently, methodological advancements have addressed many known limitations in cRSA. For example, cross-validated distance measures (e.g., Euclidean distance) have improved the reliability of representational dissimilarities in the presence of noise and trial imbalance (Walther et al., 2016; Nili et al., 2014; Diedrichsen et al., 2021). Bayesian approaches such as pattern component modeling (Diedrichsen, Yokoi, & Arbuckle, 2018) have extended representational approaches to accommodate continuous stimulus features or temporal variation. Further, model comparison RSA strategies (Diedrichsen et al., 2021) and generalization techniques across stimuli (Schütt et al., 2023) have improved sensitivity and inference. Nevertheless, a common feature shared across most of improvements is that they require stimuli repetition to examine the representational structure. This requirement limits their ability to probe brain-behavior questions at the level of individual events”.

Page 8. “While several extensions of RSA have addressed key limitations in noise sensitivity, stimulus variance, and modeling (e.g., Diedrichsen et al., 2021; Schütt et al., 2023), our tRSA approach introduces a new methodological step by estimating representational strength at the trial level. This accounts for the multi-level variance structure in the data, affords generalizability beyond the fixed stimulus set, and allows one to test stimulus- or trial-level modulations of neural representations in a straightforward way”.

Page 44. “Despite such prevalent appreciation for the neurocognitive relevance of stimulus properties, cRSA often does not account for the fact that the same stimulus (e.g., “basketball”) is seen by multiple subjects and produces statistically dependent data, an issue addressed by Schütt et al., 2023, who developed cross validation and bootstrap methods that explicitly model dependence across both subjects and stimulus conditions”.

> *(3) The stated problem of the paper is to estimate "representational strength" in different regions or conditions. With this, the authors define the correlation of the brain RDM with a model RDM. This metric conflates a number of factors, namely the variances of the stimulus-specific patterns, the variance of the noise, the true differences between different dissimilarities, and the match between the assumed model and the data-generating model. It took me a long time to figure out that the authors are trying to solve a quite different problem in a quite different setting from the model-comparative approach to RSA that I would consider "classical" (Diedrichsen et al. 2021; Diedrichsen and Kriegeskorte 2017). In this approach, one is trying to test whether local activity patterns are better explained by representation model A or model B, and to estimate the degree to which the representation can be fully explained. In this framework, it is common practice to measure each stimulus at least 2 times, to be able to estimate the variance of noise patterns and the variance of signal patterns directly. Using this setting, I would define 'representational strength" very differently from the authors. Assume (using LaTeX notation) that the activity patterns y_j,n for stimulus j, measurement n, are composed of a true stimulus-related pattern (u_j) and a trial-specific noise pattern (e_j,n). As a measure of the strength of representation (or pattern), I would use an unbiased estimate of the variance of the true stimulus-specific patterns across voxels and stimuli (σ^2_{u}). This estimator can be obtained by correlating patterns of the same stimuli across repeated measures, or equivalently, by averaging the cross-validated Euclidean distances (or with spatial prewhitening, Mahalanobis distances) across all stimulus pairs. In contrast, the current paper addresses a specific problem in a quite specific experimental design in which there is only one repetition per stimulus. This means that the authors have no direct way of distinguishing true stimulus patterns from noise processes. The trick that the authors apply here is to assume that the brain data comes from the assumed model RDM (a somewhat sketchy assumption IMO) and that everything that reduces this correlation must be measurement noise. I can now see why*

> *tRSA does make some sense for this particular question in this memory study. However, in the more common model-comparative RSA setting, having only one repetition per stimulus in the experiment would be quite a fatal design flaw. Thus, the paper would do better if the authors could spell the specific problem addressed by their method right in the beginning, rather than trying to set up tRSA as a general alternative to "classical RSA".*

At a general level, our approach rests on the premise that there is meaningful information present in a single presentation of a given stimulus. This assumption may have less utility when the research goals are more focused on estimating the fidelity of signal patterns for RSA, as in designs with multiple repetitions. But it is an exaggeration to state that such a trial-wise approach cannot address the difference between "true" stimulus patterns and noise. This trial-wise approach has explicit utility in relating trial-wise brain information to trial-wise behavior, across multiple cognitions (not only memory studies, as applied here). We have added substantial text to the Introduction distinguishing cRSA, which is widely employed, often in cases with a single repetition per stimulus, and model comparative methods that employ multiple repetitions. We clarify that we do not consider tRSA an alternative to the model comparative approach, and discuss that operational definitions of representational strength are constrained by the study design.

Page 3. "In this paper, we present an advancement termed trial-level RSA, or tRSA, which addresses these limitations in cRSA (not model comparison approaches) and may be utilized in paradigms with or without repeated stimuli".

Page 4. "Representational geometry usually refers to the structure of similarities among repeated presentations of the same stimulus in the neural data (as captured in the brain RSM) and is often estimated utilizing a model comparison approach, whereas representational strength is a derived measure that quantifies how strongly this geometry aligns with a hypothesized model RSM. In other words, geometry characterizes the pattern space itself, while representational strength reflects the degree of correspondence between that space and the theoretical model under test".

Finally, we clarified that in our simulation methods we assume a true underlying activity pattern and a random error pattern. The model RSM is computed based on the true pattern, whereas the brain RSM comes from the noisy pattern, not the model RSM itself.

Page 9. "Then, we generated two sets of noise patterns, which were controlled by parameters σ_A and σ_B , respectively, one for each condition".

> *(4) The notation in the paper is often conflicting and should be clarified. The actual true and measured activity patterns should receive a unique notation that is distinct from the variances of these patterns across voxels. I assume that σ_ijk is the noise variances (not standard deviation)? Normally, variances are denoted with σ^2. Also, if these are variances, they cannot come from a normal distribution as indicated on page 10. Finally, multi-level models are usually defined at the level of means (i.e., patterns) rather than at the level of variances (as they seem to be done here).*

We have added notations for true and measured activity patterns to differentiate it from our notation for variance. We agree that multilevel models are usually defined at the level of means rather than at the level of variances and we include a Figure (Fig 1D) that describes the model in terms of the means. We clarify that the σ (σ) used in the manuscript were not variances/standard deviations themselves; rather, they were meant to denote components of the actual (multilevel) variance parameter. Each component was sampled from normal distributions, and they collectively summed up to comprise the final variance parameter for each trial. We have modified our notation for each component to the lowercase letter s to minimize confusion. We have also made our R code publicly available on our lab github, which should provide more clarity on the exact simulation process.

> *(5) In the first set of simulations, the authors sampled both model and brain RSM by drawing each cell (similarity) of the matrix from an independent bivariate normal distribution. As the authors note themselves, this way of producing RSMs violates the constraint that correlation matrices need to be positive semi-definite. Likely more seriously, it also ignores the fact that the different elements of the upper triangular part of a correlation matrix are not independent from each other (Diedrichsen et al. 2021). Therefore, it is not clear that this simulation is close enough to reality to provide any valuable insight and should be removed from the paper, along with the extensive discussion about why this simulation setting is plainly wrong (page 21). This would shorten and clarify the paper.*

We have added justification of the mixed-effects model given the potential assumption violations. We caution readers to investigate the robustness of their models, and to employ permutation testing that does not make independence assumptions. We have also added checks of the model residuals and an example of permutation testing in the Appendix. Finally, we agree that the first simulation setting does not possess several properties of realistic RDMs/RSMs; however, we believe that there is utility in understanding the mathematical properties of correlations – an essential component of RSA – in a straightforward simulation where the ground truth is known, thus moving the simulation to Appendix 1.

> *(6) If I understand the second simulation setting correctly, the true pattern for each stimulus was generated as an NxP matrix of i.i.d. standard normal variables. Thus, there is no condition-specific pattern at all, only condition-specific noise/signal variances. It is not clear how the tRSA would be biased if there were a condition-specific pattern (which, in reality, there usually is). Because of the i.i.d. assumption of the true signal, the correlations between all stimulus pairs within conditions are close to zero (and only differ from it by the fact that you are using a finite number of voxels). If you added a condition-specific pattern, the across-condition RSA would lead to much higher "representational strength" estimates than a within-condition RSA, with obvious problems and biases.*

The Reviewer is correct that the voxel values in the true pattern are drawn from i.i.d. standard normal distributions. We take the Reviewer's suggestion of "condition-specific pattern" to mean that there could be a condition-voxel interaction in two non-mutually exclusive ways. The first is additive, essentially some common underlying multi-voxel pattern like [6, 34, -52, ..., 8] for all condition A trials, and different one such pattern for condition B trials, etc. The second is multiplicative, essentially a vector of scaling factors [x1.5, x0.5, x0.8, ..., x2.7] for all condition A trials, and a different one such vector for condition B trials, etc. Both possibilities could indeed affect tRSA as much as it would cRSA.

Importantly, If such a strong condition-specific pattern is expected, one can build a condition-specific model RDM using one-shot coding of conditions (see example figure; src: *https://www.newbi4fmri.com/tutorial-9-mvpa-rsa*), to either capture this interesting phenomenon or to remove this out as a confounding factor. This practice has been applied in multiple regression cRSA approaches (e.g., Cichy et al., 2013) and can also be applied to tRSA.

> *(7) The trial-level brain RDM to model Spearman correlations was analyzed using a mixed effects model. However, given the symmetry of the RDM, the correlations coming from different rows of the matrix are not independent, which is an assumption of the mixed effect model. This does not seem to induce an increase in Type I errors in the conditions studied, but there is no clear justification for this procedure, which needs to be justified.*

We appreciate this important warning, and now caution readers to investigate the robustness of their models, and consider employing permutation testing that does not make independence assumptions. We have also added checks of the model residuals and an example of permutation testing in the supplement.

Page 46. “While linear mixed-effects modeling offers a powerful framework for analyzing representational similarity data, it is critical that researchers carefully construct and validate their models. The multilevel structure of RSA data introduces potential dependencies across subjects, stimuli, and trials, which can violate assumptions of independence if not properly modeled. In the present study, we used a model that included random intercepts for both subjects and stimuli, which accounts for variance at these levels and improves the generalizability of fixed-effect estimates. Still, there is a potential for systematic dependence across trials within a subject. To ensure that the model assumptions were satisfied, we conducted a series of diagnostic checks on an exemplar ROI (right LOC; middle occipital gyrus) in the Object Perception dataset, including visual inspection of residual distributions and autocorrelation (Appendix 3, Figure 13). These diagnostics supported the assumptions of normality, homoscedasticity, and conditional independence of residuals. In addition, we conducted permutation-based inference, similar to prior improvements to cRSA (Niliet al. 2014), using a nested model comparison to test whether the mean similarity in this ROI was significantly greater than zero. The observed likelihood ratio test statistic fell in the extreme tail of the null distribution (Appendix 3, Figure 14), providing strong nonparametric evidence for the reliability of the observed effect. We emphasize that this type of model checking and permutation testing is not merely confirmatory but can help validate key assumptions in RSA modeling, especially when applying mixed-effects models to neural similarity data. Researchers are encouraged to adopt similar procedures to ensure the robustness and interpretability of their findings”.

Exemplar Permutation Testing

To test whether the mean representational strength in the ROI right LOC (middle occipital gyrus) was significantly greater than zero, we used a permutation-based likelihood ratio test implemented via the permlmer function. This test compares two nested linear mixed-effects models fit using the lmer function from the lme4 package, both including random intercepts for Participant and Stimulus ID to account for between-subject and between-item variability.

The null model excluded a fixed intercept term, effectively constraining the mean similarity to zero after accounting for random effects:

```
ROI ~ 0 + (1 | Participant) + (1 | Stimulus)
```

The full model included the same random effects structure but allowed the intercept to be freely estimated:

```
ROI ~ 1 + (1 | Participant) + (1 | Stimulus)
```

By comparing the fit of these two models, we directly tested whether the average similarity in this ROI was significantly different from zero. Permutation testing (1,000 permutations) was used to generate a nonparametric p-value, providing inference without relying on normality assumptions. The full model, which estimated a nonzero mean similarity in the right LOC (middle occipital gyrus), showed a significantly better fit to the data than the null model that fixed the mean at zero ($\chi^2(1) = 17.60$, $p = 2.72 \times 10^{-5}$). The permutation-based p-value obtained from permlmer confirmed this effect as statistically significant ($p = 0.0099$), indicating that the mean similarity in this ROI was reliably greater than zero. These results support the conclusion that the right LOC contains representational structure consistent with the HMAXc2 RSM. A density plot of the permuted likelihood ratio tests is plotted along with the observed likelihood ratio test in Appendix 3 Figure 14.

> *(8) For the empirical data, it is not clear to me to what degree the "representational strength" of cRSA and tRSA is actually comparable. In cRSA, the Spearman correlation assesses whether the distances in the data RSM are ranked in the same order as in the model. For tRSA, the comparison is made for every row of the RSM, which introduces a larger degree of flexibility (possibly explaining the higher correlations in the first simulation). Thus, could the gains presented in Figure 7D not simply arise from the fact that you are testing different questions? A clearer theoretical analysis of the difference between the average row-wise Spearman correlation and the matrix-wise Spearman correlation is urgently needed. The behavior will likely vary with the structure of the true model RDM/RSM.*

We agree that the comparability between mean row-wise Spearman correlations and the matrix-wise Spearman correlation is needed. We believe that the simulations are the best approach for this comparison, since they are much more robust than the empirical dataset and have the advantage of knowing the true pattern/noise levels. We expand on our comparison of mean tRSA values and matrix-wise Spearman correlations on page 42.

Page 42. "Although tRSA and cRSA both aim to quantify representational strength, they differ in how they operationalize this concept. cRSA summarizes the correspondence between RSMs as a single measure, such as the matrix-wise Spearman correlation. In contrast, tRSA computes such correspondence for each trial, enabling estimates at the level of individual observations. This flexibility allows trial-level variability to be modeled directly, but also introduces subtle differences in what is being measured. Nonetheless, our simulations showed that, although numerical differences occasionally emerged—particularly when comparing between-condition tRSA estimates to within-condition cRSA estimates—the magnitude of divergence was small and did not affect the outcome of downstream statistical tests".

> *(9) For the real data, there are a number of additional sources of bias that need to be considered for the analysis. What if there are not only condition-specific differences in noise variance, but also a condition-specific pattern? Given that the stimuli were measured in 3 different imaging runs, you cannot assume that all measurement noise is i.i.d. - stimuli from the same run will likely have a higher correlation with each other.*

We recognize the potential of condition-specific patterns and chose to constrain the analyses to those most comparable with cRSA. However, depending on their hypotheses, researchers may consider testing condition RSMs and utilizing a model comparison approach or employ the z-scored approach, as employed in the simulations above. Regarding the potential run confounds, this is always the case in RSA and why we exclude within-run comparisons. We have also added to the Discussion the suggestion to include run as a covariate in their mixed-effects models. However, we do not employ this covariate here as we preferred the most parsimonious model to compare with cRSA.

Page 46 - 47. "Further, while analyses here were largely employed to be comparable with cRSA, researchers should consider taking advantage of the flexibility of the mixed-effects models and include co variates of non-interest (run, trial order etc.)".

> *(10) The discussion should be rewritten in light of the fact that the setting considered here is very different from the model-comparative RSA in which one usually has multiple measurements per stimulus per subject. In this setting, existing approaches such as RSA or PCM do indeed allow for the full modelling of differences in the "representational strength" - i.e., pattern variance across subjects, conditions, and stimuli.*

We agree that studies advancing designs with multiple repetitions of a given stimulus image are useful in estimating the reliability of concept representations. We would argue however that model comparison in RSA is not restricted to such data. Many extant studies do not in fact have multiple repetitions per stimulus per subject (Wang et al., 2018 *https://doi.org/10.1088/1741-2552/abecc3*, Gao et al, 2022 *https://doi.org/10.1093/cercor/bhac058*, Li et al, 2022 *https://doi.org/10.1002/hbm.26195*, Staples & Graves, 2020 *https://doi.org/10.1162/nol_a_00018*) that allow for that type of model-comparative approach. While beneficial in terms of noise estimation, having multiple presentations was not a requirement for implementing cRSA (Kriegeskorte, 2008 *https://doi.org/10.3389/neuro.06.004.2008*). The aim of this manuscript is to introduce the tRSA approach to the broad community of researchers whose research questions and datasets could vary vastly, including but not limited to the number of repeated presentations and the balance of trial counts across conditions.

> *(11) Cross-validated distances provide a powerful tool to control for differences in measurement noise variances and possible covariances in measurement noise across trials, which has many distinct advantages and is conceptually very different from the approach taken here.*

We have added language on the value of cross-validation approaches to RSA in the Discussion:

Page 47. “Additionally, we note that while our proposed tRSA framework provides a flexible and statistically principled approach for modeling trial-level representational strength, we acknowledge that there are alternative methods for addressing trial-level variability in RSA. In particular, the use of cross-validated distance metrics (e.g., crossnobis distance) has become increasingly popular for controlling differences in measurement noise variance and accounting for possible covariance structures across trials (Walther et al., 2016). These metrics offer several advantages, including unbiased estimation of representational dissimilarities under Gaussian noise assumptions and improved generalization to unseen data. However, cross-validated distances are conceptually distinct from the approach taken here: whereas cross-validation aims to correct for noise-related biases in representational dissimilarity matrices, our trial-level RSA method focuses on estimating and modeling the variability in representation strength across individual trials using mixed-effects modeling. Rather than proposing a replacement for cross-validated RSA, tRSA adds a complementary tool to the methodological toolkit—one that supports hypothesis-driven inference about condition effects and trial-level covariates, while leveraging the full structure of the data”.

> *(12) One of the main limitations of tRSA is the assumption that the model RDM is actually the true brain RDM, which may not be the case. Thus, in theory, there could be a different model RDM, in which representational strength measures would be very different. These differences should be explained more fully, hopefully leading to a more accessible paper.*

Indeed, the chosen model RSM may not be the true RSM, but as the noise level increases the correlation between RSMs practically becomes zero. In our simulations we assume this to be true as a straightforward way to manipulate the correspondence between the brain data and the model. However, just like cRSA, tRSA is constrained by the model selections the researchers employ. We encourage researchers to have carefully considered theoretically-motivated models and, if their research questions require, consider multiple and potentially competing models. Furthermore, the trial-wise estimates produced by tRSA encourage testing competing models within the multiple regression framework. We have added this language to the Discussion.

Page 46. ..”choose their model RSMs carefully. In our simulations, we designed our model RSM to be the “true” RSM for demonstration purposes. However, researchers should consider if their models and model alternatives”.

Pages 45-46. “While a number of studies have addressed the validity of measuring representational geometry using designs with multiple repetitions, a conceptual benefit of the tRSA approach is the reliance on a regression framework that engenders the testing of competing conceptual models of stimulus representation (e.g., taxonomic vs. encyclopedic semantic features, as in Davis et al., 2021)".

> ***Reviewer #2 (Public review):***
>
> *(1) While I generally welcome the contribution, I take some issue with the accusatory tone of the manuscript in the Introduction. The text there (using words such as 'ignored variances', 'errouneous inferences', 'one must', 'not well-suited', 'misleading') appears aimed at turning cRSA in a 'straw man' with many limitations that other researchers have not recognized but that the new proposed method supposedly resolves. This can be written in a more nuanced, constructive manner without accusing the numerous users of this popular method of ignorance.*

We apologize for the unintended accusatory tone. We have clarified the many robust approaches to RSA and have made our Introduction and Discussion more nuanced throughout (see also 3, 11 and16).

> *(2) The described limitations are also not entirely correct, in my view: for example, statistical inference in cRSA is not always done using classic parametric statistics such as t-tests (cf Figure 1): the rsatoolbox paper by Nili et al. (2014) outlines non-parametric alternatives based on permutation tests, bootstrapping and sign tests, which are commonly used in the field. Nor has RSA ever been conducted at the row/column level (here referred to by the authors as 'trial level'; cf King et al., 2018).*

We agree there are numerous methods that go beyond cRSA addressing these limitations and have added discussion of them into our manuscript as well as an example analysis implementing permutation tests on tRSA data (see response to 7). We thank the reviewer for bringing King et al., 2014 and their temporal generalization method to our attention, we added reference to acknowledge their decoding-based temporal generalization approach.

Page 8. "It is also important to note that some prior work has examined similarly fine-grained representations in time-resolved neuroimaging data, such as the temporal generalization method introduced by King et al. (see King & Dehaene, 2014). Their approach trains classifiers at each time point and tests them across all others, resulting in a temporal generalization matrix that reflects decoding accuracy over time. While such matrices share some structural similarity with RSMs, they do not involve correlating trial-level pattern vectors with model RSMs nor do their second-level models include trial-wise, subject-wise, and item-wise variability simultaneously".

> *(3) One of the advantages of cRSA is its simplicity. Adding linear mixed effects modeling to RSA introduces a host of additional 'analysis parameters' pertaining to the choice of the model setup (random effects, fixed effects, interactions, what error terms to use) - how should future users of tRSA navigate this?*

We appreciate the opportunity to offer more specific proscriptions for those employing a tRSA technique, and have added them to the Discussion:

Page 46. "While linear mixed-effects modeling offers a powerful framework for analyzing representational similarity data, it is critical that researchers carefully construct and validate their models and choose their model RSMs carefully. In our simulations, we designed our model RSM to be the "true" RSM for demonstration purposes. However, researchers should consider if their models and model alternatives. However, researchers should always consider if their models match the goals of their analysis, including 1) constructing the random effects structure that will converge in their dataset and 2) testing their model fits against alternative structures (Meteyard & Davies, 2020; Park et al., 2020) and 3) considering which effects should be considered random or fixed depending on their research question".

> *(4) Here, only a single real fMRI dataset is used with a quite complicated experimental design for the memory part; it's not clear if there is any benefit of using tRSA on a simpler real dataset. What's the benefit of tRSA in classic RSA datasets (e.g., Kriegeskorte et al., 2008), with fixed stimulus conditions and no behavior?*

To clarify, our empirical approach uses two different tasks: an Object Perception task more akin to the classic RSA datasets employing passive viewing, and a Conceptual Retrieval task that more directly addresses the benefits of the trialwise approach. We felt that our Object Perception dataset is a simpler empirical fMRI dataset without explicit task conditions or a dichotomous behavioral outcome, whereas the Retrieval dataset is more involved (though old/new recognition is the most common form of memory retrieval testing) and dependent on behavioral outcomes. However, we recognize the utility of replication from other research groups and do invite researchers to utilize tRSA on their datasets.

> *(5) The cells of an RDM/RSM reflect pairwise comparisons between response patterns (typically a brain but can be any system; cf Sucholutsky et al., 2023). Because the response patterns are repeatedly compared, the cells of this matrix are not independent of one another. Does this raise issues with the validity of the linear mixed effects model? Does it assume the observations are linearly independent?*

We recognize the potential danger for not meeting model assumptions. Though our simulation results and model checks suggest this is not a fatal flaw in the model design, we caution readers to investigate the robustness of their models, and consider employing permutation testing that does not make independence assumptions. We have also added checks of the model residuals and an example of permutation testing in the Appendix. See response to R1.

> *(6) The manuscript assumes the reader is familiar with technical statistical terms such as Type I/II error, sensitivity, specificity, homoscedasticity assumptions, as well as linear mixed models (fixed effects, random effects, etc). I am concerned that this jargon makes the paper difficult to understand for a broad readership or even researchers currently using cRSA that might be interested in trying tRSA.*

We agree this jargon may cause the paper to be difficult to understand. We have expanded/added definitions to these terms throughout the methods and results sections.

Page 12. "Given data generated with $s_{cond,A} = s_{cond,\mathrm{B}}$, the correct inference should be a failure to reject the null hypothesis of ; any significant () result in either direction was considered a false positive (spurious effect, or Type I error). Given data generated with , the inference was considered correct if it rejected the null hypothesis of and yielded the expected sign of the estimated contrast ($b_{\mathrm{B}\text{-}A}<0$). A significant result with the reverse sign of the estimated contrast ($b_{\mathrm{B}\text{-}A}<0$) was considered a Type I error, and a nonsignificant ($p \geq 0.05$) result was considered a false negative (failure to detect a true effect, or Type II error)".

Page 2. "Compared to cRSA, the multi-level framework of tRSA was both more theoretically appropriate and significantly sensitive (better able to detect) to true effects".

Page 25."The performance of cRSA and tRSA were quantified with their specificity (better avoids false positives, 1 - Type I error rate) and sensitivity (better avoids false negatives 1 - Type II error rate)".

Page 6. "One of the fundamental assumptions of general linear models (step 4 of cRSA; see Figure 1D) is homoscedasticity or homogeneity of variance — that is, all residuals should have equal variance" .

Page11. “Specifically, a linear mixed-effects model with a fixed effect of condition (which estimates the average effect across the entire sample, capturing the overall effect of interest) and random effects of both subjects and stimuli (which model variation in responses due to differences between individual subjects and items, allowing generalization beyond the sample) were fitted to tRSA estimates via the ̀lme4 1.1-35.3 ̀package in R (Bates et al., 2015), and p-values were estimated using Satterthwaites’s method via the ̀lmerTest 3.1-3 ̀package (Kuznetsova et al., 2017)”.

> *(7) I could not find any statement on data availability or code availability. Given that the manuscript reuses prior data and proposes a new method, making data and code/tutorials openly available would greatly enhance the potential impact and utility for the community.*

We thank the reviewer for raising our oversight here. We have added our code and data availability statements.

Page 9. “Data is available upon request to the corresponding author and our simulations and example tRSA code is available at *https://github.com/electricdinolab*”.

> ***Reviewer #1 (Recommendations for the authors):***
>
> *(13) Page 4: The limitations of cRSA seem to be based on the assumption that within each different experimental condition, there are different stimuli, which get combined into the condition. The framework of RSA, however, does not dictate whether you calculate a condition x condition RDM or a larger and more complete stimulus x stimulus RDM. Indeed, in practice we often do the latter? Or are you assuming that each stimulus is only shown once overall? It would be useful at this point to spell out these implicit assumptions.*

We agree that stimulus x stimulus RDMs can be constructed and are often used. However, as we mentioned in the Introduction, researchers are often interested in the difference between two (or more) conditions, such as “remembered” vs. “forgotten” (Davis et al., *https://doi.org/10.1093/cercor/bhaa269*) or “high cognitive load” vs. “low cognitive load” (Beynel et al., *https://doi.org/10.1523/JNEUROSCI.0531-20.2020*). In those cases, the most common practice with cRSA is to construct condition-specific RDMs, compute cRSA scores separately for each condition, and then compare the scores at the group level. The number of times each stimulus gets presented does not prevent one from creating a model RDM that has the same rows and columns as the brain RDM, either in the same condition (“high load”) or across different conditions.

> *(14) Page 5: The difference between condition-level and stimulus-level is not clear. Indeed, this definition seems to be a function of the exact experimental design and is certainly up for interpretation. For example, if I conduct a study looking at the activity patterns for 4 different hand actions, each repeated multiple times, are these actions considered stimuli or conditions?*

We have added clarifying language about what is considered stimuli vs conditions. Indeed, this will depend on the specific research questions being employed and will affect how researchers construct their models. In this specific example, one would most likely consider each different hand action a condition, treating them as fixed effects rather than random effects, given their very limited number and the lack of need to generalize findings to the broader “hand actions” category.

Page 5. “Critically, the distinction between condition-level and stimulus level is not always clear as researchers may manipulate stimulus-level features themselves. In these cases, what researchers ultimately consider condition-level and stimulus-level will depend on their specific research questions. For example, researchers intending to study generalized object representation may consider object category a stimulus-level feature, while researchers interested in if/how object representation varies by category may consider the same category variable condition-level".

> *(15) Page 5: The fact that different numbers of trials / different levels of measurement noise / noise-covariance of different conditions biases non-cross-validated distances is well known and repeatedly expressed in the literature. We have shown that cross-validation of distances effectively removes such biases - of course, it does not remove the increased estimation variability of these distances (for a formal analysis of estimation noise on condition patterns and variance of the cross-nobis estimator, see (Diedrichsen et al. 2021)).*

We thank the reviewer for drawing our attention to this literature and have added discussions of these methods.

> *(16). Page 5: "Most studies present subjects with a fixed set of stimuli, which are supposedly samples representative of some broader category". This may be the case for a certain type of RSA experiments in the visual domain, but it would be unfair to say that this is a feature of RSA studies in general. In most studies I have been involved in, we use a "stimulus" x "stimulus" RDM.*

We have edited this sentence to avoid the "most" characterization. We also added substantial text to the introduction and discussion distinguishing cRSA, which is nonetheless widely employed, especially in cases with a single repetition per stimulus (Macklin et al., 2023, Liu et al, 2024) and the model comparative method and explicitly stating that we do not consider tRSA an alternative to the model comparative approach.

> *(17). Page 5: I agree that "stimuli" should ideally be considered a random effect if "stimuli" can be thought of as sampled from a larger population and one wants to make inferences about that larger population. Sometimes stimuli/conditions are more appropriately considered a fixed effect (for example, when studying the response to stimulation of the 5 fingers of the right hand). Techniques to consider stimuli/conditions as a random effect have been published by the group of Niko Kriegeskorte (Schütt et al. 2023).*

Indeed, in some cases what may be thought of as "stimuli" would be more appropriately entered into the model as a fixed effect; such questions are increasingly relevant given the focus on item-wise stimulus properties (Bainbridge et al., Westfall & Yarkoni). We have added text on this issue to the Discussion and caution researchers to employ models that most directly answer their research questions.

Page 46. "However, researchers should always consider if their models match the goals of their analysis, including 1) constructing the random effects structure that will converge in their dataset and 2) testing their model fits against alternative structures (Meteyard & Davies, 2020; Park et al., 2020) and 3) considering which effects should be considered random or fixed depending on their research question. An effect is fixed when the levels represent the specific conditions of theoretical interest (e.g., task condition) and the goal is to estimate and interpret those differences directly. In contrast, an effect is random when the levels are sampled from a broader population (e.g., subjects) and the goal is to account for their variability while generalizing beyond the sample tested. Note that the same variable (e.g., stimuli) may be considered fixed or random depending on the research questions".

> *(18) Page 6: It is correct that the "classical" RSA depends on a categorical assignment of different trials to different stimuli/conditions, such that a stimulus x stimulus RDM can be computed. However, both Pattern Component Modelling (PCM) and Encoding models are ideally set up to deal with variables that vary continuously on a trial-by-trial or moment-*

> *by-moment basis. tRSA should be compared to these approaches, or - as it should be clarified - that the problem setting is actually quite a different one.*

We agree that PCM and encoding models offer a flexible approach and handle continuous trial-by-trial variables. We have clarified the problem setting in cRSA is distinct on page 6, and we have added the robustness of encoding models and their limitations to the Discussion.

Page 6. "While other approaches such as Pattern Component Modeling (PCM) (Diedrichsen et al., 2018) and encoding models (Naselaris et al., 2011) are well-suited to analyzing variables that vary continuously on a trial-by-trial or moment-by-moment basis, these frameworks address different inferential goals. Specifically, PCM and encoding models focus on estimating variance components or predicting activation from features, while cRSA is designed to evaluate representational geometry. Thus, cRSA as well as our proposed approach address a problem setting distinct from PCM and encoding models".

> *(19) Page 8: "Then, we generated two noise patterns, which were controlled by parameters σA and σB, respectively, one for each condition." This makes little sense to me. The noise patterns should be unique to each trial - you should generate n_a + n_b noise patterns, no?*

We clarify that the "noise patterns" here are n_voxel x n_trial in size; in other words, all trial-level noise patterns are generated together and each trial has their own unique noise pattern. We have revised our description as "two sets of noise patterns" for clarity starting on page 9.

> *(20) Page 9: First, I assume if this is supposed to be a hierarchical level model, the "noise parameters" here correspond to variances? Or do these \sigma values mean to signify standard deviations? The latter would make little sense. Or is it the noise pattern itself?*

As clarified in 4., the σ values are meant to denote hierarchical components of the composite standard deviation; we have updated our notation to use lower case letter s instead for clarity.

> *(21) Page 10: your formula states "σ_{subj}~ N(0, 0.5^2)". This conflicts with your previous mention that \sigmas are noise "levels" are they the noise patterns themselves now? Variances cannot be normally distributed, as they cannot be negative.*

As clarified in 4., the σ values are meant to denote hierarchical components of the composite standard deviation; we have updated our notation to use lower case letter s instead for clarity.

> *(22) Page 13: What was the task of the subject in the Memory retrieval task? Old/new judgements relative to encoding of object perception?*

We apologize for the lack of clarity about the Memory Retrieval task and have added that information and clarified that the old/new judgements were relative to a separate encoding phase, the brain data for which has been reported elsewhere.

Page 14. "Memory Retrieval took place one day after Memory Encoding and involved testing participants' memory of the objects seen in the Encoding phase. Neural data during the Encoding phase has been reported elsewhere. In the main Memory Retrieval task, participants were presented with 144 labels of real-world objects, of which 114 were labels for previously seen objects and 30 were unrelated novel distractors. Participants performed old/new judgements, as well as their confidence in those judgements on a four-point scale (1 = Definitely New, 2 = Probably New, 3 = Probably Old, 4 = Definitely Old)".

> *(23) Page 13: If "Memory Retrieval consisted of three scanning runs", then some of the stimulus x stimulus correlations for the RSM must have been calculated within a run and some between runs, correct? Given that all within-run estimates share a common*

> *baseline, they share some dependence. Was there a systematic difference between the within-run and the between-run correlations?*

We have clarified in this portion of the methods that within run comparisons were excluded from our analyses. We also double-checked that the within-run exclusion was included in the description of the Neural RSMs.

Page 14. “Retrieval consisted of three scanning runs, each with 38 trials, lasting approximately 9 minutes and 12 seconds (within-run comparisons were later excluded from RSA analyses)”.

Page 18. “This was done by vectorizing the voxel-level activation values within each region and calculating their correlations using Pearson’s r, excluding all within-run comparisons.”

> *(24) Page 20: It is not clear why the mean estimate of "representational strength" (i.e., model-brain RSM correlations) is important at all. This comes back to Major point #2, namely that you are trying to solve a very different problem from model-comparative RSA.*

We have clarified that our approach is not an alternative to model-comparative RSA, and that depending on the task constraints researchers may choose to compare models with tRSA or other approaches requiring stimulus repetition (see 3).

> *(25) Page 21: I believe the problems of simulating correlation matrices directly in the way that the authors in their first simulation did should be well known and should be moved to an appendix at best. Better yet, the authors could start with the correct simulation right away.*

We agree the paper is more concise with these simulations being moved to the appendix and more briefly discussed. We have implemented these changes (Appendix 1). However, we are not certain that this problem is unknown, and have several anecdotes of researchers inquiring about this “alternative” approach in talks with colleagues, thus we do still discuss the issues with this method.

> *(26) Page 26: Is the "underlying continuous noise variable $\sigma trial$ that was measured by $v measured$ " the variance of the noise pattern or the noise pattern itself? What does it mean it was "measured" - how?*

$\sigma trial$ is a vector of standard deviations for different trials, and $\sigma trial$ i would be used to generate the noise patterns for trial i. v_measured is a hypothetical measurement of trial-level variability, such as “memorability” or “heartbeat variability”. We have revised our description to clarify our methods.

> ***Reviewer #2 (Recommendations for the authors):***
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> *(8) It would be helpful to provide more clarity earlier on in the manuscript on what is a 'trial': in my experience, a row or column of the RDM is usually referred to as 'stimulus condition', which is typically estimated on multiple trials (instances or repeats) of that stimulus condition (or exemplars from that stimulus class) being presented to the subject. Here, a 'trial' is both one measurement (i.e., single, individual presentation of a stimulus) and also an entry in the RDM, but is this the most typical scenario for cRSA? There is a section in the Discussion that discusses repetitions, but I would welcome more clarity on this from the get-go.*

We have added discussion of stimulus repetition methods and datasets to the Introduction and clarified our use of the terms.

Page 8. “Critically, in single-presentation designs, a “trial” refers to one stimulus presentation, and corresponds to a row or column in the RSM. In studies with repeated stimuli, these rows are often called “conditions” and may reflect aggregated patterns across trials. tRSA is compatible with both cases: whether rows represent individual trials or averaged trials that create “conditions”, tRSA estimates are computed at the row level”.

> *(9) The quality of the results figures can be improved. For example, axes labels are hard to read in Figure 3A/B, panels 3C/D are hard to read in general. In Figure 7E, it's not possible to identify the 'dark red' brain regions in addition to the light red ones.*

We thank the reviewer for raising these and have edited the figures to be more readable in the manner suggested.

> *(10) I would be interested to see a comparison between tRSA and cRSA in other fMRI (or other modality) datasets that have been extensively reported in the literature. These could be the original Kriegeskorte 96 stimulus monkey/fMRI datasets, commonly used open datasets in visual perception (e.g., THINGS, NSD), or the above-mentioned King et al. dataset, which has been analyzed in various papers.*

We recognize the great utility of replication from other research groups and do invite researchers to utilize tRSA on their datasets.

> *(11) On P39, the authors suggest 'researchers can confidently replace their existing cRSA analysis with tRSA': Please discuss/comment on how researchers should navigate the choice of modeling parameters in tRSA's linear mixed effects setting.*

We have added discussion of the mixed-effects parameters and the various and encourage researchers to follow best practices for their model selection.

Page 46. “However, researchers should always consider if their models match the goals of their analysis, including 1) constructing the random effects structure that will converge in their dataset and 2) testing their model fits against alternative structures (Meteyard & Davies, 2020; Park et al., 2020) and 3) considering which effects should be considered random or fixed depending on their research question”.

> *(12) The final part of the Results section, demonstrating the tRSA results for the continuous memorability factor in the real fMRI data, could benefit from some substantiation/elaboration. It wasn't clear to me, for example, to what extent the observed significant association between representational strength and item memorability in this dataset is to be 'believed'; the Discussion section (p38). Was there any evidence in the original paper for this association? Or do we just assume this is likely true in the brain, based on prior literature by e.g. Bainbridge et al (who probably did not use tRSA but rather classic methods)?*

Indeed, memorability effects have been replicated in the literature, but not using the tRSA method. We have expanded our discussion to clarify the relationship of our findings and the relevant literature and methods it has employed.

Page 38. “Critically, memorability is a robust stimulus property that is consistent across participants and paradigms (Bainbridge, 2022). Moreover, object memorability effects have been replicated using a variety of methods aside from tRSA, including univariate analyses and representational analyses of neural activity patterns where trial-level neural activity pattern estimates are correlated directly with object memorability (Slayton et al, 2025).”

(13) The abstract could benefit from more nuance; I'm not sure if RSA can indeed be said to be 'the principal method', and whether it's about assessing 'quality' of representations (more commonly, the term 'geometry' or 'structure' is used).

We have edited the abstract to reflect the true nuisance in the current approaches.

Abstract. Neural representation refers to the brain activity that stands in for one's cognitive experience, and in cognitive neuroscience, a prominent method of studying neural representations is representational similarity analysis (RSA). While there are several recent advances in RSA, the classic RSA (cRSA) approach examines the structure of representations across numerous items by assessing the correspondence between two representational similarity matrices (RSMs): usually one based on a theoretical model of stimulus similarity and the other based on similarity in measured neural data.

(14) RSA is also not necessarily about models vs. neural data; it can also be between two neural systems (e.g., monkey vs. human as in Kriegeskorte et al., 2008) or model systems (see Sucholutsky et al., 2023). This statement is also repeated in the Introduction paragraph 1 (later on, it is correctly stated that comparing brain vs. model is most likely the 'most common' approach).

We have added these examples in our introduction to RSA.

Page 3."One of the central approaches for evaluating information represented in the brain is representational similarity analysis (RSA), an analytical approach that queries the representational geometry of the brain in terms of its alignment with the representational geometry of some cognitive model (Kriegeskorte et al., 2008; Kriegeskorte & Kievit, 2013), or, in some cases, compares the representational geometry of two neural systems (e.g., Kriegeskorte et al., 2008) or two model systems (Sucholutsky et al., 2023)".

> *(15) 'theoretically appropriate' is an ambiguous statement, appropriate for what theory?*

We apologize for the ambiguous wording, and have corrected the text:

Page 11. "Critically, tRSA estimates were submitted to a mixed-effects model which is statistically appropriate for modeling the hierarchical structure of the data, where observations are nested within both subjects and stimuli (Baayen et al., 2008; Chen et al., 2021)".

> *(16) I found the statement that cRSA "cannot model representation at the level of individual trials" confusing, as it made me think, what prohibits one from creating an RDM based on single-trial responses? Later on, I understood that what the authors are trying to say here (I think) is that cRSA cannot weigh the contributions of individual rows/columns to the overall representational strength differently.*

We thank the reviewer for their clarifying language and have added it to this section of the manuscript.

"Abstract. However, because cRSA cannot weigh the contributions of individual trials (RSM rows/columns), it is fundamentally limited in its ability to assess subject-, stimulus-, and trial-level variances that all influence representation".

> *(17) Why use "RSM" instead of "RDM"? If the pairwise comparison metric is distance-based (e..g, 1-correlation as described by the authors), RDM is more appropriate.*

We apologize for the error, and have clarified the Methods text:

Page3-4. First, brain activity responses to a series of N trials are compared against each other (typically using Pearson's r) to form an N×N representational similarity matrix.

> *(18) Figure 2: please write 'Correlation estimate' in the y-axis label rather than 'Estimate'.*

We have edited the label in Figure 2.

> *(19) Page 6 'leaving uncertain the directionality of any findings' - I do not follow this argument. Obviously one can generate an RDM or RSM from vector v or vector -v. How does that invalidate drawing conclusions where one e.g., partials out the (dis)similarity in e.g., pleasantness ratings out of another RDM/RSM of interest?*

We agree such an approach does not invalidate the partial method; we have clarified what we mean by "directionality".

Page 8. "For instance, even though a univariate random variable , such as pleasantness ratings, can be conveniently converted to an RSM using pairwise distance metrics (Weaverdyck et al., 2020), the very same RSM would also be derived from the opposite random variable , leaving uncertain of the directionality (or if representation is strongest for pleasant or unpleasant items) of any findings with the RSM (see also Bainbridge & Rissman, 2018)".

> *(20) P7 'sampled 19900 pairs of values from a bi-variate normal distribution', but the rows/columns in an RDM are not independent samples - shouldn't this be included in the simulation? I.e., shouldn't you simulate first the n=200 vectors, and then draw samples from those, as in the next analysis?*

This section has been moved to Appendix 1 (see responses to Reviewer 1.13).

> *(21) Under data acquisition, please state explicitly that the paper is re-using data from prior experiments, rather than collecting data anew for validating tRSA.*

We have clarified this in the data acquisition section.

Page 13. "A pre-existing dataset was analyzed to evaluate tRSA. Main study findings have been reported elsewhere (S. Huang, Bogdan, et al., 2024)".

> *(22) Figure 4 could benefit from some more explanation in-text. It wasn't clear to me, for example, how to interpret the asterisks depicted in the right part of the figure.*

We clarified the meaning of the asterisks in the main text in addition to the existent text in the figure caption.

Page 26. "see Figure 4, off-diagonal cells in blue; asterisks indicate where tRSA was statistically more sensitive then cRSA)".

> *(23) Page 38 "the outcome of tRSA's improved characterization can be seen in multiple empirical outcomes:" it seems there is one mention of 'outcomes' too many here.*

We have revised this sentence.

Page 41. "tRSA's improved characterization can be seen in multiple empirical outcomes".

> *(24) Page 38 "model fits became the strongest" it's not clear what aspect of the reported results in the paragraph before this is referring to - the Appendix?*

Yes, the model fits are in the Appendix, we have added this in text citation.

Moreover, model-fits became the strongest when the models also incorporated trial-level variables such as fMRI run and reaction time (Appendix 3, Table 6).

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